# Groups, Rings and Modules

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In this course we will continue our study of groups, focusing on simple groups, $p$-groups and $p$-subgroups along with the Sylow theorems. We will then move on to discuss rings and fields, finishing up by discussing modules where the scalars belong to a ring instead of a field.

This article constitutes my notes for the 'Groups, Rings and Modules' course, held in Lent 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

## Contents

# §1 The Theory of Groups

## §1.1 Definition of a Group

We will begin by recalling the basic definition of a group.

**Definition 1.1** (Group)

A **group** is a pair $(G, *)$ consisting of a set $G$ and a binary operation $* : G \times G \to G$ satisfying the axioms:

- *Identity.* There is an element $e \in G$ such that $e * g = g * e = g$ for all $g \in G$,
- *Inverses.* For every element $g \in G$, there is an element $g^{-1} \in G$ such that $g * g^{-1} = g^{-1} * g = e$.

- *Associativity.* The operation $*$ is associative.

**Remark.** We will usually either use additive or multiplicative notation for groups, and in these cases we will often write 0 or 1 for the identity respectively.

**Definition 1.2** (Subgroup)
A subset $H \subseteq G$ is a **subgroup** of $G$, written $H \leq G$, if it is a group with respect to the operation $*$ defined on $H \times H$.

There is a way to test the conditions needed for a subset to be a subgroup in just a few lines, but it does have limited utility.

**Lemma 1.3** (Fast Subgroup Checking)
A nonempty subset $H \subseteq G$ is a subgroup if $a, b \in H$ implies $a * b^{-1} \in H$.

*Proof Sketch.* Check that this implies the definition. □

**Example 1.4** (Examples of Groups)
The following are all examples of groups.

(i) The additive groups $(\mathbb{Z}, +) \leq (\mathbb{Q}, +) \leq (\mathbb{R}, +)$.

(ii) The cyclic group of order $n$, $C_n$.

(iii) The dihedral group $D_{2n}$ of the symmetries of a regular $n$-gon.

(iv) The symmetric group $S_n$ and alternating group $A_n$, where $S_n$ is the group of permutations of $\{1, 2, \dots, n\}$ and $A_n \leq S_m$ is the group of even permutations.

(v) The quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ with $i^2 = j^2 = k^2 = ijk = -1$.

(vi) The matrix groups over some field $F$, $\mathrm{GL}_n(F)$ of all $n \times n$ matrices over $F$ with non-zero determinant, and $\mathrm{SL}_n(F) \leq \mathrm{GL}_n(F)$, the subgroup of matrices with determinant 1.

**Definition 1.5** (Direct Product)
The **direct product** of groups $G$ and $H$ is $G \times H$ with operation $(g_1, h_1) * (g_2, h_2) = (g_1 g_2, h_1 h_2)$.

## §1.2 Lagrange's Theorem

For a subgroup $H \leq G$, the **left cosets** of $H$ in $G$ are the sets $gH = \{gh \mid h \in H\}$ where $g \in G$. The left cosets partition $G$, and it is straightforward to show that each has the same cardinality as $H$. These facts immediately give us *Lagrange's theorem.*

**Theorem 1.6** (Lagrange's Theorem)

If $G$ is a finite group and $H \leq G$, then

$$|G| = |H| \cdot |G : H|,$$

where $|G : H|$ is the **index** of $H$ in $G$, the number of left cosets of $H$ in $G$.

**Remark.** The proof of Lagrange's theorem is equally valid when using right cosets, and as a consequence there is always the same number of left and right cosets for a given subgroup.

**Definition 1.7** (Order of an Element)

Let $G$ be a group and $g \in G$. The least $n$ such that $g^n = 1$ is the **order** of $g$. If no such $n$ exists, we say that $g$ has **infinite order**.

**Remark.** If $g$ has order $d$, then the division algorithm implies $g^n = 1 \iff d \mid n$. Also by taking powers we can generate a cyclic subgroup of $G$ with $\{1, g, g^2, \ldots, g^{d-1}\} \leq G$, so if $G$ is finite then by Lagrange, $d \mid |G|$.

## §1.3 Homomorphisms and Isomorphisms

As with many mathematical objects, there are many instances in which we are interested in the structure preserving maps between groups. Such maps are called *homomorphisms*.

**Definition 1.8** (Homomorphism)

If $G$ and $H$ are groups, a function $\phi : G \to H$ is a **group homomorphism** if $\phi(g_1 g_2) = \phi(g_1)\phi(g_2)$ for all $g_1, g_2 \in G$.

We frequently care about what elements in the group $H$ can be obtained through this map, and also what elements in $G$ just get sent to the identity.

**Definition 1.9** (Kernel)

The **kernel** of $\phi$ is $\ker(\phi) = \{g \in G \mid \phi(g) = e\}$.

**Definition 1.10** (Image)

The **image** of $\phi$ is $\operatorname{img}(\phi) = \{\phi(g) \mid g \in G\}$.

It is straightforward to see that $\ker(\phi) \leq G$ and $\operatorname{img}(\phi) \leq H$.

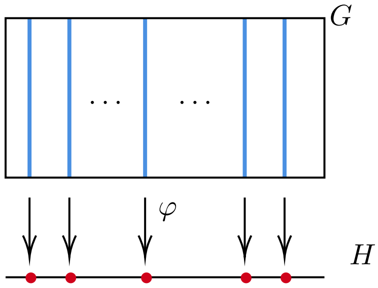
When we have a homomorphism that gives a bijection between two groups, the groups must have the same underlying algebraic structure. For this reason, we take this to be a notion of two groups being the same.

**Definition 1.11** (Isomorphism)

We say that two groups $G$ and $H$ are **isomorphic** written $G \cong H$ if there is an *isomorphism* between them. That is, if there is a bijective homomorphism $\phi : G \to H$ between the groups.

## §1.4 Quotient Groups and the Isomorphism Theorems

Quotient groups arise naturally in the study of homomorphisms. When we have two groups $G$ and $H$, and some homomorphism $\varphi : G \to H$, we can see that the elements of $G$ mapping to certain elements of $H$ naturally arrange themselves together.




Indeed, if two elements of $G$ map to the same element of $H$ under $\varphi$, we would find that the two elements must differ by an element of the kernel of $\varphi$. So, going back to how the elements of $G$ arrange themselves, it would seem that they are arranged into cosets of the kernel of $\varphi$. It is this observation that leads us to the definition of quotient groups and the isomorphism theorems.

From the discussion above, it is natural to see when we can make the cosets of a given subgroup into another group. It turns out that this only works when we have a specific type of subgroup.

**Definition 1.12** (Normal Subgroup)

We say that a subgroup $N$ of a group $G$ is **normal** if the left and right cosets coincide, that is, $gN = Ng$ for all $g \in G$. If $N$ is normal in $G$ we write $N \trianglelefteq G$.

Now we can show that the set of cosets of $N$ in $G$, written $G/N$, forms a group.

**Theorem 1.13** (Quotient Groups)

If $N \trianglelefteq G$, then the set of cosets of $N$ in $G$ forms the **quotient group** $G/N$ with the operation of coset multiplication, where $gN \cdot g'N = (gg')N$.

*Proof.* We first need to check that coset multiplication is well defined. If we had $a_1N = a_2N$ and $b_1N = b_2N$, then $a_1 = a_2n_a$ and $b_1 = b_2n_b$ for some $n_a, n_b \in N$. Then $(a_1b_1)N = (a_2n_ab_2n_b)N = (a_2b_2)N$, since $N$ is normal in $G$. So our group operation is well defined, and we are just left to check the group axioms.

We have closure since $g_1N$ and $g_2N$ being cosets implies $(g_1g_2)N$ is too. Then we have the identity element $N$, and inverses with $gN \cdot g^{-1}N = N$. Finally we get associativity inherited from $G$, thus we do indeed have a group. □

Now we initially tried to motivate thinking about groups of cosets by noticing that elements in the same coset of the kernel of $\phi$ had the same image. We then used this 'normal' property of a subgroup to make this actually work. It turns out that these express the same idea – normal subgroups are *exactly* the kernels of homomorphisms.

**Theorem 1.14** (Kernels are Normal Subgroups)

If $\phi : G \to H$ is a homomorphism, then $\ker \phi \trianglelefteq G$.

*Proof.* We already know that the kernel of a homomorphism is a subgroup, so we just need to check that it is normal. Let $n \in \ker \phi$, and $g \in G$. Then $\phi(gng^{-1}) = \phi(g)\phi(n)\phi(g^{-1}) = \phi(g)\phi(g^{-1}) = e$, thus $gng^{-1} \in \ker \phi$, so $\ker \phi \trianglelefteq G$ as required. □

**Theorem 1.15** (Normal Subgroups are Kernels)
Given $N \trianglelefteq G$, the map $\pi : G \to G/N$ with $\pi(g) = gN$ is a surjective homomorphism called the **quotient map**, and $\ker \pi = N$.

*Proof.* We first check that $\pi$ is a homomorphism. For $g, h \in G$, we have $\pi(gh) = (gh)N = gN \cdot hN = \pi(g)\pi(h)$ as required. This map is clearly surjective, and also $\pi(g) = gN = N \iff g \in N$, so $\ker \pi = N$, as required. □

Looking back again to our original motivation, we can see that the cosets of the kernel seem to form the group that is the image of the homomorphism. This is the exact idea encompassed in quotient groups, and is formalised in the *first isomorphism theorem*.

**Theorem 1.16** (First Isomorphism Theorem)
Let $\phi : G \to H$ be a group homomorphism. Then $\ker(\phi) \trianglelefteq G$, and $G/\ker(\phi) \cong \operatorname{img}(\phi)$.

*Proof.* Let $K = \ker(\phi)$. We already checked that $K \trianglelefteq G$. Now define $\Phi : G/K \to \operatorname{img}(\phi)$ by $gK \mapsto \phi(g)$.

We first check $\Phi$ is well defined and injective. We have

$$\begin{aligned} g_1K = g_2K &\iff g_2^{-1}g_1 \in K \\ &\iff \phi(g_2^{-1}g_1) = e \\ &\iff \phi(g_2)^{-1}\phi(g_1) = e \\ &\iff \phi(g_1) = \phi(g_2). \end{aligned}$$

Then we check that $\Phi$ is a group homomorphism, with

$$\begin{aligned} \Phi(g_1Kg_2K) &= \Phi(g_1g_2K) \\ &= \phi(g_1g_2) \\ &= \phi(g_1)\phi(g_2) \\ &= \Phi(g_1K)\Phi(g_2K). \end{aligned}$$

Lastly we check that it is surjective. Let $x \in \operatorname{img}(\phi)$, say $x = \phi(g)$ for some $g \in G$. Then $x = \Phi(gK) \in \operatorname{img}(\Phi)$. □

In a sense, this is the way that you should think about normal subgroups and quotient groups – as the kernels and images of homomorphisms.

**Example 1.17** (Using the First Isomorphism Theorem)

Let $\phi : \mathbb{C} \to \mathbb{C}^*$ with $z \mapsto e^z$. As $e^{z+w} = e^z e^w$, this is a group homomorphism from $(\mathbb{C}, +)$ to $(\mathbb{C}^*, \times)$.

We find that $\ker(\phi) = \{z \in \mathbb{C} \mid e^z = 1\} = 2\pi i\mathbb{Z}$, and $\operatorname{img}(\phi) = \mathbb{C}^*$. Thus $\mathbb{C}/2\pi i\mathbb{Z} \cong \mathbb{C}^*$.

With the first isomorphism theorem, it is not enough to know the statement and proof – you have to know when to employ it. For example, if asked to prove $\mathbb{C}/2\pi i\mathbb{Z} \cong \mathbb{C}^*$, you should be able to think of a strategy similar to the one used above.

The first isomorphism theorem is sometimes just called the 'isomorphism theorem', and it tends to be more important than the corollaries that we will state.

**Corollary 1.18** (Second Isomorphism Theorem)

Let $H \leq G$ and $K \trianglelefteq G$. Then $HK = \{hk \mid h \in H, k \in K\} \leq G$ and $H \cap K \trianglelefteq H$, moreover $HK/K \cong H/H \cap K$.

*Proof.* Let $h_1k_1, h_2k_2 \in HK$. We have

$$\begin{aligned} h_1k_1(h_2k_2)^{-1} &= h_1k_1k_2^{-1}h_2^{-1} \\ &= h_1h_2^{-1}h_2k_1k_2^{-1}h_2^{-1} \\ &= (h_1h_2^{-1})(h_2k_1k_2^{-1}h_2^{-1}) \in HK, \end{aligned}$$

thus $HK \leq G$, as required.

Now let $\phi : H \mapsto G/K$ with $h \mapsto hK$. This is the composite of the inclusion $H \hookrightarrow G$ and the quotient map $G \to G/K$, thus $\phi$ is a group homomorphism.

We note $\ker(\phi) = \{h \in H \mid hK = k\} = H \cap K \trianglelefteq H$, and $\operatorname{img}(\phi) = \{hK \mid h \in H\} = HK/K$. Thus by the first isomorphism theorem, $H/H \cap K \cong HK/K$. $\square$

Before we state the third isomorphism theorem, consider the following motivation. Suppose $K \trianglelefteq G$. There is a bijection

$$\{\text{subgroups of } G/K\} \longleftrightarrow \{\text{subgroups of } G \text{ containing } K\},$$

obtained by considering the maps $X \mapsto \{g \in G \mid gK \in X\}$ and $H \mapsto H/K$. This restricts to a bijection between

$$\{\text{normal subgroups of } G/K\} \longleftrightarrow \{\text{normal subgroups of } G \text{ containing } K\}.$$

In pictures, the portion of the subgroup lattice of $G$ sitting above $K$ matches the entire subgroup lattice of $G/K$.

$$\begin{tikzcd} G \arrow[rr, leftrightarrow, dashed] \arrow[d, dash] & & G/K \arrow[d, dash] \\ H \arrow[rr, leftrightarrow, dashed] \arrow[d, dash] & & H/K \arrow[d, dash] \\ K \arrow[rr, leftrightarrow, dashed] \arrow[d, dash] & & \{e\} \\ \{e\} & & \end{tikzcd}$$

**Corollary 1.19** (Third Isomorphism Theorem)
Let $K \leq H \leq G$ be normal subgroups of $G$. Then

$$(G/K)/(H/K) \cong G/H.$$

*Proof.* Let $\phi : G/K \to G/H$ with $gK \mapsto gH$. If $g_1K = g_2K$, then $g_2^{-1}g_1 \in K \leq H$, so $g_1H = g_2H$, and thus $\phi$ is well defined. Also $\phi$ is a surjective group homomorphism with kernel $\ker(\phi) = H/K$. Then apply the first isomorphism. □

## §1.5 Simple Groups

If $K \trianglelefteq G$, then studying the groups $K$ and the quotient group $G/K$ gives some information about $G$. However, this approach is not always available.

**Definition 1.20** (Simple Group)
A group $G$ is **simple** if $\{e\}$ and $G$ are its only normal subgroups.

**Lemma 1.21** (Abelian Simple Groups)
An abelian group is simple if and only if it is isomorphic to $C_p$ for some prime $p$.

*Proof.* By Lagrange's theorem, a subgroup $H \leq C_p$ has order dividing $|C_p| = p$, which is a prime. Hence $H$ has order 1 or $p$, and $H$ is either $\{e\}$ or $H = G$.

Now let $G$ be an abelian simple group, and $g \in G$ with $g \neq e$. Note that any subgroup of an abelian group is normal, and thus $G$ must have no subgroups other than $G$ and $\{e\}$. But then $G$ has subgroup $\langle g \rangle = \{\dots, g^{-2}, g^{-1}, e, g, g^2, \dots\}$. Since $G$ is simple, this must be the whole group, that is, $G$ is cyclic.

If $G$ is the infinite cyclic group, then $G \cong (\mathbb{Z}, +)$, which is not simple (as $2\mathbb{Z} \trianglelefteq \mathbb{Z}$). Thus $G \cong C_n$ for some $n$. Let $g$ be a generator for $C_n$. If $m \mid n$, then $\langle g^{n/m} \rangle$ is a subgroup of order $m$, but $G$ is simple thus $m = 1$ or $n$, hence $n$ must be prime. □

**Lemma 1.22** (Composition Series of Finite Groups)
If $G$ is a finite group then $G$ has a composition series $\{e\} = G_0 \triangleleft G_1 \triangleleft G_2 \triangleleft \cdots \triangleleft G_{m-1} \triangleleft G_m = G$, with each quotient $G_i/G_{i-1}$ is simple.

[Note that $G_i$ need not be normal in $G$.]

*Proof.* We induct on $|G|$. If $|G| = 1$ we are done. If $|G| > 1$, then let $G_{m-1}$ be a normal subgroup of largest possible order (not $|G|$). Then $G/G_{m-1}$ is simple, and by induction on $G_{m-1}$, we are done. □

# §2 Group Actions

A useful way to study a group is by studying how it 'acts' on some set. The way we look at this mathematically is through the lens of group actions.

## §2.1 Definitions

We will begin by looking at groups of permutations of a set.

**Definition 2.1** (Sym($X$))

For a set $X$, let $\mathrm{Sym}(X)$ be the group of all bijections $X \to X$ under composition. We let the identity of this group be id.

**Definition 2.2** (Permutation Group)

A group $G$ is a **permutation group** (of degree $n$) if $G \leq \mathrm{Sym}(X)$ (where $|X| = n$).

**Example 2.3** (Examples of Permutation Groups)

The group $S_n = \mathrm{Sym}(\{1, 2, \ldots, n\})$ is a permutation group of degree $n$, as is the alternating group $A_n \leq S_n$.

The group $D_n$, the symmetries of a regular $n$-gon, is a permutation group as it is a subgroup of Sym({vertices of an $n$-gon}).

We can now generalize the notion of a permutation group to the idea mentioned before – slightly more useful (and general) than a permutation group is the notion of a group *acting* on a set.

**Definition 2.4** (Group Action)

An action of a group $G$ on a set $X$ is a function $* : G \times X \to X$ satisfying

(i) $e * x = x$ for all $x \in X$.

(ii) $(g_1 g_2) * x = g_1 * (g_2 * x)$ for all $g_1, g_2 \in G$ and $x \in X$.

We can also think about group actions in the following way, and switching between the two points of view can be quite helpful.

**Proposition 2.5**

An action of a group $G$ on a set $X$ is equivalent to specifying a group homomorphism $\phi : G \to \mathrm{Sym}(X)$.

*Proof.* For each $g \in G$, there is a function $\phi_g : X \to X$ given by $x \mapsto g * x$. We have $\phi_{g_1 g_2}(x) = (g_1 g_2) * x = g_1 * (g_2 * x) = \phi_{g_1}(\phi_{g_2}(x))$. Thus $\phi_{g_1 g_2} = \phi_{g_1} \circ \phi_{g2}$.

In particular, $\phi_g \circ \phi_{g^{-1}} = \phi_{g^{-1}} \circ \phi_g = \phi_e = \mathrm{id}$. Thus $\phi_g$ is a bijection, and $\phi_g \in \mathrm{Sym}(X)$. We define $\phi : G \to \mathrm{Sym}(X)$ with $g \mapsto \phi_g$. Then this is a group homomorphism by the above.

Conversely, let $\phi : G \to \mathrm{Sym}(X)$ be a group homomorphism. Then defining $* : G \times X \to X$ with $(g, x) \mapsto \phi(g)(x)$, we have that this is a group action since $e * x = \phi(e)(x) = \mathrm{id}(x) = x$ and $(g_1 g_2) * x = \phi(g_1 g_2)(x) = \phi(g_1)(\phi(g_2)(x)) = g_1 * (g_2 * x)$. □

**Definition 2.6** (Permutation Representation)

We say $\phi : G \to \mathrm{Sym}(X)$ is a **permutation representation** of $G$.

## §2.2 Orbits and Stabilisers

A useful notion is that of *orbits* and *stabilisers*. Informally, the orbit of an element $x$ in a set $S$ acted on by a group $G$ is all of the elements that can be reached by applying elements of $G$. The stabiliser of $x$ is the set of elements in $G$ so that when we apply them, we still have the element $x$.

**Definition 2.7** (Orbits and Stabilisers)

Let $G$ act on a set $X$. The **orbit** of an element $x \in X$ is $\mathrm{Orb}_G(x) = \{g * x \mid g \in G\}$, which is a subset of $X$. The **stabiliser** of $x \in X$ is $\mathrm{Stab}_G(x) = \{g \in G \mid g * x = x\}$, which is a subgroup of $G$.

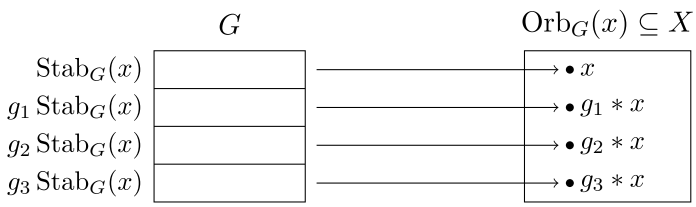
The orbits partition the set $X$. If there is only one orbit then we say that the group action is **transitive**. We recall the Orbit-Stabiliser theorem.

**Theorem 2.8** (Orbit-Stabiliser Theorem)

There is a bijection between $\mathrm{Orb}_G(x)$ and the set of left cosets of $\mathrm{Stab}_G(x)$ in $G$. In particular, if $G$ is finite, then

$$|G| = |\operatorname{Orb}_G(x)| \cdot |\operatorname{Stab}_G(x)|.$$

The picture to have in mind is the following: the cosets of $\mathrm{Stab}_G(x)$ partition $G$, and all of the elements of a given coset $g \,\mathrm{Stab}_G(x)$ move $x$ to the *same* point $g * x$ of the orbit, with distinct cosets giving distinct points.




**Remark.** The kernel of $\phi$, which can be written as $\ker \phi = \bigcap_{x \in X} \mathrm{Stab}_G(x)$, is called the **kernel of the group action**. Also $\mathrm{Stab}(g * x) = g \,\mathrm{Stab}(x) g^{-1}$, so if $x, y \in X$ belong to the same orbit then their stabilisers are conjugate subgroups of $G$.

## §2.3 Using Group Actions

In many cases, choosing the right group action can help make progress on a problem. With this in mind, it's helpful to have a list of standard group actions that you can apply.

**Example 2.9** (Examples of Group Actions)

The following are all group actions.

(i) Let $G$ act on itself by left multiplication, with $g * x = gx$. The kernel of this action is $\{g \in G \mid gx = x\} = \{e\}$, and thus $G$ injects into $\mathrm{Sym}(G)$. This proves Cayley's theorem, that any finite group $G$ is isomorphic to a subgroup of $S_n$ for some $n$ (say $n = |G|$).

(ii) Let $H \leq G$. Then $G$ acts on the left cosets of $H$ in $G$ by left multiplication. This group action is transitive, with $\mathrm{Stab}_G(xH) = \{g \in G \mid gxH = xH\} = xHx^{-1}$. We have the kernel $\ker(\phi) = \bigcap_{x \in G} xHx^{-1}$, which is the largest normal subgroup of $G$ contained in $H$.

(iii) Let $G$ act on itself by conjugation, so $g * x = gxg^{-1}$. Here, the orbits and stabilisers are $\mathrm{Orb}_G(x) = \{gxg^{-1} \mid g \in G\} = \mathrm{ccl}_G(x)$, the **conjugacy class** of $x$ in $G$. The stabilisers are $\mathrm{Stab}_G(x) = \{g \in G \mid gx = xg\} = C_G(x) \leq G$, the **centraliser** of $x$. The kernel of this action is the **center**, $Z(G) = \{g \in G \mid gx = xg\ \forall x \in G\}$.

$G$ also acts by conjugation on any normal subgroup.

(iv) Let $X$ be the set of all subgroups of $G$. Then $G$ acts on $X$ by conjugation. That is, $g * H = gHg^{-1} \leq G$. The stabiliser of $H$ is $\{g \in G \mid gHg^{-1} = H\} = N_G(H)$ is the **normaliser** of $H$ in $G$. This is the largest subgroup of $G$ to contain $H$ as a normal subgroup. In particular $H \trianglelefteq G \iff N_G(H) = G$.

Of this last, the frequently useful actions (that you should readily reach for) are the left multiplication of cosets, conjugation of elements and subgroups.

To see how we can apply group actions to proving a theorem, consider the following result (and mainly it's proof). Here we will use the action of $G$ on the left cosets of a subgroup $H$.

**Theorem 2.10**

Let $G$ be a non-abelian simple group, and $H \leq G$ a subgroup of index $n > 1$ in $G$. Then $n \geq 5$ and $G$ is isomorphic to a subgroup of $A_n$.

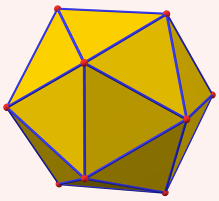
*Proof.* Let $G$ act on $X$, the set of left cosets of $H$ in $G$, by left multiplication, and let $\phi : G \to \mathrm{Sym}(X)$ be the associated permutation representation. Then $\mathrm{Sym}(X) = S_n$. As $G$ is simple, we know that $\ker \phi = \{e\}$ or $G$. If $\ker \phi = G$, then $\operatorname{img} \phi = \{e\}$, contradicting that $G$ acts transitively on $X$ (since $n > 1$). Thus $\ker \phi$ is trivial, and $G \cong \operatorname{img} \phi \leq S_n$.

Since $G \leq S_n$ and $A_n \trianglelefteq S_n$, the second isomorphism theorem gives $G \cap A_n \trianglelefteq G$ and $G/G \cap A_n \leq S_n/A_n \cong C_2$. But $G$ is simple so $G \cap A_n = \{e\}$ or $G$. If $G \cap A_n = \{e\}$, then $G \hookrightarrow C_2$, which contradicts that $G$ is non-abelian. Otherwise, $G \cap A_n = G$, and $G \leq A_n$. Finally if $n \leq 4$, then $A_n$ has no non-abelian simple subgroups (by listing them), so $n \geq 5$. $\square$

Another example, looking at the rotational symmetries of a icosahedron, is given below.

**Example 2.11** (Rotational Symmetries of an Icosahedron)

Let $G$ be the group of rotations of an icosahedron, as pictured.



In this shape, there is 20 faces, 12 vertices and 30 edges. We want to know the possible orders of elements in this group, and the number of elements of each order.

| Order | Number of Elements in G |
|---|---|
| 1 | 1 |
| 2 | 15 |
| 3 | 20 |
| 5 | 24 |

This gives us a total of 60 elements. We can verify this with the orbit-stabiliser theorem. For $G$ acting on the vertices, and some vertex $v$, we have $|G| = |\operatorname{Orb}(v)| \cdot |\operatorname{Stab}(x)| = 12 \cdot 5 = 60$.

To consider the conjugacy classes, we note that two elements are conjugate if they 'look the same up to relabelling the icosahedron'.

The elements of order 2 are all conjugate, as are those by order 3. The elements of order 5 split into two conjugacy classes of equal size (12), as the rotations by $\pm\frac{2\pi}{5}$ are all conjugate and rotations by $\pm\frac{4\pi}{5}$ are all conjugate, but not to each-other.

We can use this to deduce $G$ is simple. If $H \trianglelefteq G$, then $|H| = 1 + 15a + 20b + 12c$ for $a, b \in \{0, 1\}$ and $c \in \{0, 1, 2\}$, but as $|H| \mid 60$ by Lagrange, we get $|H| = 1$ or 60. Thus $G$ is simple.

In fact, $G$ is isomorphic to $A_5$. We will show that the sets $H \setminus \{1\}$ for $H \leq G$ a subgroup of order 4 partition the 15 elements of order 2 into 5 sets of 3.

1. If $|H| = 4$, then $H \cong C_2 \times C_2$ or $H \cong C_4$. But there is no elements of order 4 in $G$, so $H \cong C_2 \times C_2$. Note that this has three elements of order 2.
2. If $g \in G$ has order 2, then $g \in C_G(g)$ and $|C_G(g)| = \frac{|G|}{|\operatorname{ccl}_G(g)|} = \frac{60}{15} = 4$.
3. Suppose $e \neq g \in H \cap K$ where $H$ and $K$ are distinct subgroups of order 4. Then $|C_G(g)| \geq |H \cup K|$ since $H$ and $K$ are abelian, and $|H \cup K| > 4$.

This proves the claim. Now let $G$ act on the set $X$ of subgroups of order 4 by conjugation. We obtain a group homomorphism $\phi : G \to \operatorname{Sym}(X) \cong S_5$. Then $G$ simple implies either $\ker \phi = \{e\}$ or $G$. If it was $G$, then we would get that there is a normal subgroup of order 4, which is a contradiction. Thus the kernel is trivial, and $G \hookrightarrow S_5$. Then exactly as before, either $G \hookrightarrow C_2$ or $G \leq A_5$, and thus $|G| = |A_5| = 60$, so $G \cong A_5$.

# §3 Alternating Groups

We know from the 'Groups' course that two permutations in $S_n$ are conjugate if and only if they have the same cycle type.

**Example 3.1** (Conjugacy Classes in $S_5$)
In $S_5$, we have the following:

| Cycle Type | Number of Elements | Sign |
|---|---|---|
| $1,1,1,1,1$ | 1 | $+$ |
| $2,1,1,1$ | 10 | $-$ |
| $2,2,1$ | 15 | $+$ |
| $3,1,1$ | 20 | $+$ |
| $3,2$ | 20 | $-$ |
| $4,1$ | 30 | $-$ |
| $5$ | 24 | $+$ |

Thus we know the sizes of the conjugacy classes in $S_5$. The same method works in general for $S_n$.

We want to think about the conjugacy classes of $A_n$.

## §3.1 Conjugacy Classes & Simplicity of $A_n$

One way to think about conjugacy classes is to say that the group acts on itself by conjugation. Then the conjugacy classes correspond to orbits and centralizers correspond to stabilisers.

So it's going to be useful to think about the centralisers of elements in $A_n$. Let $g \in A_n$. Then $C_{A_n}(g) = C_{S_n}(g) \cap A_n$. We have then got two cases.

- If there exists an odd permutation commuting with $g$, then $|C_{A_n}(g)| = \frac{1}{2}|C_{S_n}(g)|$, and $|\operatorname{ccl}_{A_n}(g)| = |\operatorname{ccl}_{S_n}(g)|$.
- Otherwise, $|C_{A_n}(g)| = |C_{S_n}(g)|$ and $|\operatorname{ccl}_{A_n}(g)| = \frac{1}{2}|\operatorname{ccl}_{S_n}(g)|$.

**Example 3.2** (Conjugacy Classes in $A_5$)
We can now investigate the conjugacy classes in $A_5$. Note that $(1\ 2)(3\ 4)$ commutes with the odd permutation $(1\ 2)$ and $(1\ 2\ 3)$ commutes with $(4\ 5)$. But if $h \in C_{S_5}(g)$ where $g = (1\ 2\ 3\ 4\ 5)$, then

$$(1\ 2\ 3\ 4\ 5) = h(1\ 2\ 3\ 4\ 5)h^{-1} = (h(1)\ h(2)\ h(3)\ h(4)\ h(5)).$$

Thus $h \in \langle g \rangle$, and $|\operatorname{ccl}_{A_5}(g)| = \frac{1}{2}|\operatorname{ccl}_{S_5}(g)| = 12$. Thus $A_5$ has conjugacy classes of size $1, 15, 20, 12, 12$. By the same argument as before, this implies that $A_5$ is simple.

**Lemma 3.3** (Generators of $A_n$)
$A_n$ is generated by 3-cycles.

*Proof.* We know that each $\sigma \in A_n$ is the product of an even number of transpositions. So it suffices to write the product of any two transpositions as the product of 3-cycles.

For $a, b, c, d$ distinct, we have $(a\ b)(b\ c) = (a\ b\ c)$, and $(a\ b)(c\ d) = (a\ c\ b)(a\ c\ d)$. □

**Lemma 3.4** (3-Cycles are Conjugate in $A_n$)
If $n \geq 5$, then all 3-cycles in $A_n$ are conjugate.

*Proof.* We claim that every 3-cycle is conjugate to $(1\ 2\ 3)$. Indeed, if $(a\ b\ c)$ is a 3-cycle, then $(a\ b\ c) = \sigma(1\ 2\ 3)\sigma^{-1}$ for some $\sigma \in S_n$. If $\sigma \notin A_n$, then replace $\sigma$ with $\sigma(4\ 5)$. □

**Theorem 3.5** ($A_n$ is Simple)
The alternating group $A_n$ is simple for all $n \geq 5$.

*Proof.* Let $N \trianglelefteq A_n$ be a non-trivial subgroup of $A_n$. It suffices to show that $N$ contains a 3-cycle.

We take some $\sigma \in N$ with $\sigma \neq e$, and write $\sigma$ as the product of disjoint cycles.

- *Case 1.* $\sigma$ contains a cycle of length $r \geq 4$.

  Without loss of generality, let $\sigma = (1\,2\,3\ \cdots\ r)\tau$. Let $\delta = (1\,2\,3)$, then $\delta^{-1}\sigma\delta \in N$, and $\sigma^{-1}\delta^{-1}\sigma\delta \in N$. This can be written $(r\ \cdots\ 2\,1)(1\,3\,2)(1\,2\ \cdots\ r)(1\,2\,3) = (2\ 3\ r)$, which is a three cycle. Then $N$ contains a three cycle.

- *Case 2.* $\sigma$ contains two 3-cycles.

  Again, without loss of generality say $\sigma = (1\ 2\ 3)(4\ 5\ 6)\tau$. Then let $\delta = (1\ 2\ 4)$, and $\sigma^{-1}\delta^{-1}\sigma\delta = (1\ 3\ 2)(4\ 6\ 5)(1\ 4\ 2)(1\ 2\ 3)(4\ 6\ 5)(1\ 2\ 3) = (1\ 2\ 4\ 3\ 6)$. Then we can apply case 1, and $N$ contains a three cycle.

- *Case 3.* $\sigma$ contains two 2-cycles.

  Without loss of generality, say $\sigma = (1\ 2)(3\ 4)\tau$. Then let $\delta = (1\ 2\ 3)$. Then $\sigma^{-1}\delta^{-1}\sigma\delta = (1\ 2)(3\ 4)(1\ 3\ 2)(1\ 2)(3\ 4)(1\ 2\ 3) = (1\ 4)(2\ 3)$. Calling this $\pi$, and letting $\epsilon = (2\,3\,5)$, then $\pi^{-1}\epsilon^{-1}\pi\epsilon = (1\ 4)(2\ 3)(2\ 5\ 3)(1\ 4)(2\ 3)(2\ 3\ 5) = (2\ 5\ 3)$, so $N$ contains a 3-cycle. Note that we use $n \geq 5$ here.

It remains to consider $\sigma$ where it is a three-cycle. But then $N$ contains a 3-cycle. □

**Definition 3.6** (Automorphism)
An **automorphism** of a group $G$ is an isomorphism $G \to G$.

The automorphisms of a group form a subgroup $\mathrm{Aut}(G) \leq \mathrm{Sym}(G)$.

# §4 $p$-Groups and $p$-Subgroups

## §4.1 Basic Properties of $p$-Groups

Throughout this section, $p$ will denote a fixed prime number.

**Definition 4.1** ($p$-group)
A finite group $G$ is a $p$-group if $|G| = p^n$ for $n \in \mathbb{N}$ where $p$ is a prime.

We will now state some basic properties of $p$-groups that will be used in the following sections.

**Theorem 4.2** ($p$-groups Have Non-trivial Centers)
If $G$ is a $p$-group, then $Z(G) \neq \{e\}$.

*Proof.* For $g \in G$, we have $|\operatorname{ccl}_G(g)| \cdot |C_G(g)| = |G| = p^n$. So each conjugacy class has size that is a power of $p$. Since $G$ is a union of its conjugacy classes,

$$\begin{aligned} |G| &\equiv \#(\text{conj. classes of size } 1) \pmod p \\ \implies 0 &\equiv |Z(G)| \pmod p. \end{aligned}$$

In particular, $|Z(G)| > 1$. □

**Corollary 4.3** (Simple $p$-groups)
The only simple $p$-group is $C_p$.

*Proof.* Let $G$ be a simple $p$-group. Since $Z(G) \trianglelefteq G$, we have either $Z(G) = \{e\}$ or $G$. The center cannot be trivial by the previous theorem, thus $Z(G) = G$, so $G$ is abelian, and the only abelian simple groups are $C_p$ for a prime $p$. □

**Corollary 4.4**
Let $G$ be a $p$-group of order $p^n$. Then $G$ has a subgroup of order $p^r$ for all $0 \leq r \leq n$.

*Proof.* We know that any finite group $G$ has a composition series $\{e\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_{m-1} \triangleleft G_m = G$, with each quotient $G_i/G_{i-1}$ simple. Then $G$ is a $p$-group implies $G_i/G_{i-1}$ is a $p$-group, and thus $G_i/G_{i-1} \cong C_p$. Then $|G_i| = p^i$ for all $0 \leq i \leq m$, and $m = n$. □

**Lemma 4.5**
For $G$ a group, if $G/Z(G)$ is cyclic, then $G$ is abelian, and $G/Z(G)$ is trivial.

*Proof.* Let $gZ(G)$ be a generator for $G/Z(G)$. Then each coset is of the form $g^r Z(G)$ for some $r \in \mathbb{Z}$. Thus $G = \{g^r z \mid r \in \mathbb{Z}, z \in Z(G)\}$. Then $(g^{r_1} z_1)(g^{r_2} z_2) = g^{r_1+r_2} z_1 z_2$ since $z_1$ is in the center. Since $z_2$ is also in the center, we can also write this as $g^{r_1+r_2}z_2z_1 = (g^{r_2}z_2)(g^{r_1}z_1)$. Thus $G$ is abelian. □

We can now start to consider what various $p$-groups actually look like. We know that a group of order $p$ is cyclic, so we will begin with groups of order $p^2$.

**Corollary 4.6** (Groups of Order $p^2$)
If $|G| = p^2$, then $G$ is abelian.

*Proof.* The center of $G$ is non-trivial, and by thus by Lagrange, $|Z(G)|$ is either $p$ or $p^2$. If the order was $p^2$, then we would be done. So consider the case $|Z(G)| = p$. Then $|G/Z(G)|$ must have order $p$, but then it is cyclic and contradicts our previous lemma. □

We will leave considering groups of order $p^3$ as an exercise.

## §4.2 Sylow Theorems

Let $G$ be a finite group, and write $|G| = p^a m$ where $p$ is a prime with $p \nmid m$.

**Theorem 4.7** (First Sylow Theorem)
The set $\mathrm{Syl}_p(G) = \{P \leq G \mid |P| = p^a\}$ of **Sylow $p$-subgroups** is non-empty.

*Proof.* We wish to show that $G$ has a subgroup of order $p^a$. Let $\Omega$ be the set of all subsets of $G$ of size $p^a$. We have

$$|\Omega| = \binom{p^a m}{p^a} = \frac{p^a m}{p^a} \cdot \frac{p^a m - 1}{p^a - 1} \cdots \frac{p^a m - p^a + 1}{1}.$$

For $0 \leq k < p^a$, the numbers $p^a m - k$ and $p^a - k$ are divisible by the same power of $p$. Thus $|\Omega|$ is coprime to $p$.

Let $G$ act on $\Omega$ by left multiplication, where for $g \in G$ and $X \in \Omega$, we put $g * X = \{gx \mid x \in X\} \in \Omega$. For any $X \in \Omega$, we have $|\operatorname{Stab}_G(X)| \cdot |\operatorname{Orb}_G(X)| = |G| = p^a m$. But by the coprimality described above, we can pick $X$ such that $|\operatorname{Orb}_G(X)|$ is coprime to $p$. Therefore $p^a \mid |\operatorname{Stab}_G(X)|$.

On the other hand, if $g \in G$ and $x \in X$, then $g \in (gx^{-1}) * X$. Therefore $G = \bigcup_{g \in G} g * X$. Thus $|G| \leq |\operatorname{Orb}_G(X)| \cdot |X|$, and thus $|\operatorname{Stab}_G(X)| = \frac{|G|}{|\operatorname{Orb}_G(X)|} \leq |X| = p^a$. So $|\operatorname{Stab}_G(X)| = p^a$, and is thus a Sylow $p$-subgroup. □

**Theorem 4.8** (Second Sylow Theorem)
All elements of $\mathrm{Syl}_p(G)$ are conjugate.

*Proof.* We claim that if $P \in \mathrm{Syl}_p(G)$ and $Q \leq G$ is a $p$-subgroup then $Q \leq gPg^{-1}$ for some $g \in G$.

Let $Q$ act on the set of left cosets of $G$ by left multiplication, where $q * gP = qgP$.

The orbit stabiliser theorem implies each orbit has size dividing $|Q|$, so it is either 1 or a multiple of $p$.

Since $|G : P| = m$ is coprime to $p$, there exists an orbit of size 1. Thus there exists $g \in G$ such that $qgP = gP$ for all $q \in Q$, and we get $g^{-1}qg \in P$, so $Q \subseteq gPg^{-1}$. $\square$

**Theorem 4.9** (Third Sylow Theorem)

The number $n_p = |\mathrm{Syl}_p(G)|$ of Sylow $p$-subgroups satisfies $n_p \equiv 1 \pmod p$, and $n_p \mid |G|$.

*Proof.* Let $G$ act on $\mathrm{Syl}_p(G)$ by conjugation. The second Sylow theorem implies that this action is transitive. So by the orbit stabiliser theorem, $n_p = |\mathrm{Syl}_p(G)|$ divides $|G|$.

Now let $P \in \mathrm{Syl}_p(G)$. $P$ acts on $\mathrm{Syl}_p(G)$ by conjugation. Then the orbits have size dividing $|P|$, so they are either 1 or a multiple of $p$.

To show $n_p \equiv 1 \pmod p$, it suffices to show that $\{P\}$ is the unique orbit of size 1. If $\{Q\}$ is an orbit of size 1, then $P$ normalizes $Q$, so $P \leq N_G(Q)$. Now $P$ and $Q$ are Sylow $p$-subgroups of $N_G(Q)$. Hence by the second Sylow theorem, they are conjugate in $N_G(Q)$, and equal since $Q \trianglelefteq N_G(Q)$. Thus $\{P\}$ is the unique orbit of size 1. $\square$

## §4.3 Applications of the Sylow Theorems

We will now look at how the Sylow theorems can be use to prove some facts about certain groups.

**Corollary 4.10**

If $n_p = 1$, then the unique Sylow $p$-subgroup is normal.

*Proof.* Let $g \in G$ and $P \in \mathrm{Syl}_p(G)$. Then $gPg^{-1} \leq G$ is another Sylow $p$-subgroup, so we must have $gPg^{-1} = P$, for all $g \in G$. Then $P \trianglelefteq G$. $\square$

**Example 4.11** (No Simple Groups of Order 1000)

Let $|G| = 1000 = 2^3 \cdot 5^3$. Then $n_5 \equiv 1 \pmod 5$, and $n_5 \mid 8$. Checking, we find that $n_5 = 1$, and the unique Sylow 5-subgroup is normal. Then $G$ is not simple.

**Example 4.12** (No Simple Groups of Order 132)

Let $|G| = 132 = 2^2 \cdot 3 \cdot 11$. Then $n_{11} \equiv 1 \pmod{11}$, and $n_{11} \mid 12$. Then $n_{11} = 1$ or 12.

Suppose $G$ was simple. Then $n_{11} \neq 1$, so $n_{11} = 12$. Now $n_3 \equiv 1 \pmod 3$, and $n_3 \mid 44$, so $n_3 = 1, 4$ or 22. But $G$ is simple, and either $n_3 = 4$ or 22. Supposing $n_3 = 4$, then letting $G$ act on $\mathrm{Syl}_3(G)$ by conjugation, we get a group homomorphism $\phi : G \to S_4$. Then $\ker(\phi) \trianglelefteq G$ implies $\ker(\phi) = \{e\}$ or $G$. It cannot be $G$ by the second Sylow theorem, thus $\phi$ is injective. But then $G \hookrightarrow S_4$, and $132 \leq 24$, which is a contradiction.

Thus $n_3 = 22$, and $n_{11} = 12$. But then counting, $G$ has $22 \cdot (3-1) = 44$ elements of order 3, and $12 \cdot (11-1) = 120$ elements of order 11. But then $120 + 44 = 164 > 132$, which is a contradiction. Thus there does not exist a simple group of order 132.

# §5 Some Groups

## §5.1 Matrix Groups

We are going to finish our study of groups by looking at some matrix groups, along with finite abelian groups.

Let $F$ be a field (for example $\mathbb{C}$ or $\mathbb{Z}/p\mathbb{Z}$). Recall the groups $\mathrm{GL}_n(F)$ of $n \times n$ invertible matrices over $F$, and $\mathrm{SL}_n(F) = \ker(\mathrm{GL}_n \xrightarrow{\det} F^\times) \trianglelefteq GL_n(F)$. Also let $Z \trianglelefteq \mathrm{GL}_n(F)$ be the group of scalar matrices (scalar multiples of the identity matrices).

We define the *projective general/special linear groups* as follows.

**Definition 5.1** (Projective General Linear Group)
The **projective general linear group** $\mathrm{PGL}_n(F)$ is defined to be $\mathrm{GL}_n(F)/Z$.

**Definition 5.2** (Projective Special Linear Group)
The **projective special linear group** $\mathrm{PSL}_n(F)$ is the group $\mathrm{SL}_n(F)/(Z \cap \mathrm{SL}_n(F))$, which is isomorphic to $Z \cdot \mathrm{SL}_n(F)/Z$, by the second isomorphism theorem.

**Example 5.3**
Let $G = \mathrm{GL}_n(\mathbb{Z}/p\mathbb{Z})$. A list of $n$ vectors in $(\mathbb{Z}/p\mathbb{Z})^n$ are the columns of some $A \in G$ if and only if they are linearly independent. Thus

$$\begin{aligned} |G| &= (p^n - 1)(p^n - p)(p^n - p^2) \cdots (p^n - p^{n-1}) \\ &= p^{1+2+\cdots+(n-1)}(p^n - 1)(p^{n-1} - 1) \cdots (p - 1) \\ &= p^{\binom{n}{2}} \prod_{i=1}^{n} (p^i - 1). \end{aligned}$$

We see that the Sylow $p$-subgroups have order $p^{\binom{n}{2}}$. One such is the subgroup of upper triangular matrices with 1's on the diagonal.

$$U = \left\{ \begin{pmatrix} 1 & * & * & \cdots & * \\ 0 & 1 & * & \cdots & * \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix} \right\} \leq G.$$

Indeed there are $\binom{n}{2}$ entries $*$, each which can take $p$ values, so the order is $p^{\binom{n}{2}}$.

Just as $\mathrm{PGL}_2(\mathbb{C})$ acts on $\mathbb{C} \cup \{\infty\}$ via Möbius maps, $\mathrm{PGL}_2(\mathbb{Z}/p\mathbb{Z})$ acts on $\mathbb{Z}/p\mathbb{Z} \cup \{\infty\}$. Indeed, $\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})$ acts as

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} : z \longmapsto \frac{az+b}{cz+d},$$

and since scalar matrices act trivially, this is an action of $\mathrm{PGL}_2(\mathbb{Z}/p\mathbb{Z})$.

**Lemma 5.4**

The permutation representation of $\mathrm{PGL}_2(\mathbb{Z}/p\mathbb{Z}) \to S_{p+1}$ is injective (and is an isomorphism if $p = 2$ or $3$).

*Proof.* Suppose $\frac{az+b}{cz+d} = z$ for all $z \in \mathbb{Z}/p\mathbb{Z} \cup \{\infty\}$ Putting $z = 0$ shows $b = 0$. Putting $z = \infty$ shows $c = 0$. Putting $z = 1$ shows $a = d$, and thus this is a scalar matrix. $\square$

**Lemma 5.5**

If $p$ is an odd prime, then $|\,\mathrm{PSL}_2(\mathbb{Z}/p\mathbb{Z})| = \frac{p(p-1)(p+1)}{2}$.

*Proof.* By the previous example, $|\,\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})| = p(p-1)(p^2-1)$. Then consider the group homomorphism $\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z}) \xrightarrow{\det} (\mathbb{Z}/p\mathbb{Z})^\times$. This is surjective, since we have

$$\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix} \longmapsto a,$$

so $|\,\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z})| = \frac{|\,\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})|}{p-1} = p(p-1)(p+1)$.

Then counting the number of scalar matrices, if

$$\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z}),$$

then $\lambda^2 \equiv 1 \pmod p$ implies $p \mid (\lambda-1)(\lambda+1)$, so $\lambda \equiv \pm 1 \pmod p$. So there is two such scalar matrices, and $|\,\mathrm{PSL}_2(\mathbb{Z}/p\mathbb{Z})| = \frac{1}{2}|\,\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z})| = \frac{p(p-1)(p+1)}{2}$. $\square$

**Example 5.6**

Let $G = \mathrm{PSL}_2(\mathbb{Z}/5\mathbb{Z})$. Then $|G| = 60 = 2^2 \cdot 3 \cdot 5$.

Let $G$ act on $\mathbb{Z}/5\mathbb{Z} \cup \{\infty\}$ via Möbius maps. By our previous lemma, there is an injective group homomorphism $\phi : G \to \mathrm{Sym}(\{0, 1, \dots, 4, \infty\}) \cong S_6$.

*Claim.* $\mathrm{img}(\phi) \le A_6$, that is, $\psi : G \to S_6 \xrightarrow{\mathrm{sgn}} \{\pm 1\}$ is trivial.

If $m$ is odd, then $\psi(g) = e \iff \psi(g)^m = e \iff \psi(g^m) = e$. So it suffices to consider $g \in G$ with order a power of 2. We showed in the proof of the Sylow theorems that every such element belongs to a Sylow 2-subgroup. So it suffices to check that $\psi(H) = e$, where $H$ is a Sylow 2-subgroup (using that we have any two Sylow 2-subgroups are conjugate, and $\psi$ maps to an abelian group).

We take $H = \left\{ \pm \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \pm \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\} \leq G$. We compute:

$$\phi\left(\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}\right) = (1\ 4)(2\ 3)$$
$$\phi\left(\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\right) = (0\ \infty)(1\ 4).$$

These are even permutations, so $\psi(H)$ is trivial, and our claim is true.

## §5.2 Finite Abelian Groups

Later in this course, we will prove the following theorem.

**Theorem 5.7** (The Fundamental Theorem of Finite Abelian Groups)
Every finite abelian group is isomorphic to the direct product of cyclic groups.

**Remark** (Non-Uniqueness)**.** While this is possible, such a decomposition is not unique. For example, you may recall from IA groups that if $m$ and $n$ are coprime, then $C_m \times C_n \cong C_{mn}$

**Lemma 5.8**
If $m$ and $n$ are coprime, then $C_{mn} = C_m \times C_n$.

*Proof.* Let $g$ and $h$ be generators of $C_m$ and $C_n$. Then we have $(g, h) \in C_m \times C_n$, and $(g, h)^r = (g^r, h^r)$. In particular, $(g, h)^r = e$ if and only if $m \mid r$ and $n \mid r$, that is, if $mn \mid r$ since they are coprime. Thus $(g, h)$ has order $mn$, which is $|C_m \times C_n|$. □

**Corollary 5.9**
Let $G$ be a finite abelian group. Then $G \cong C_{n_1} \times C_{n_2} \times \cdots \times C_{n_k}$ where each $n_i$ is a prime power.

*Proof.* If $n = p_1^{a_1} \cdots p_r^{a_r}$, with $p_i$ prime, then we can (by the previous lemma) write $C_n \cong C_{p_1^{a_1}} \times \cdots \times C_{p_r^{a_r}}$. Writing each of the cyclic groups in the fundamental theorem of abelian groups then gives the result. □

In fact, we will prove the following refinement of the the fundamental theorem of finite abelian groups.

**Theorem 5.10**
Let $G$ be a finite abelian group. Then

$$G \cong C_{d_1} \times C_{d_2} \times \cdots \times C_{d_t},$$

for some $d_1 \mid d_2 \mid \cdots \mid d_t$.

**Remark.** The integers $n_1, \ldots, n_k$ in the previous corollary and $d_1, \ldots, d_t$ in the previous theorem are uniquely determined by $G$. The proof (which we omit) works by counting the number of elements of each prime power order.

**Example 5.11** (Abelian Groups of Order 8)

The abelian groups of order 8 are $C_8$, $C_2 \times C_4$ and $C_2 \times C_2 \times C_2$.

**Example 5.12** (Abelian Groups of Order 12)

The abelian groups of order 12 are $C_2 \times C_6 \cong C_2 \times C_2 \times C_3$, $C_4 \times C_3 \cong C_{12}$.

**Definition 5.13** (Exponent)

The **exponent** of a group $G$ is the least integer $n \geq 1$ such that $g^n = e$ for all $g \in G$. That is, it is the lowest common multiple of all of the orders of elements in $G$.

**Example 5.14**

$A_4$ has exponent 6.

**Corollary 5.15**

Every finite abelian group contains an element whose order is the exponent of the group.

*Proof.* If $G \cong C_{d_1} \times \cdots \times C_{d_t}$ with $d_1 \mid d_2 \mid \cdots \mid d_t$, then every $g \in G$ has order dividing $d_t$, and if $h \in C_{d_t}$ is a generator, then $(1, 1, 1, \ldots, h) \in G$ has order $d_t$. So $G$ has exponent $d_t$. $\square$

# §6 Rings

## §6.1 Basic Definitions

We will now move on to another central algebraic object in this course – *rings*. Let's start with a definition.

**Definition 6.1** (Ring)

A **ring** is a triple $(R, +, \cdot)$ consisting of a set $R$ and two binary operations[a] $+, \cdot : R \times R \to R$, referred to as *addition* and *multiplication*, that satisfy

- $(R, +)$ is an abelian group, with identity 0.
- Multiplication is associative and has an identity 1.
- *Distributivity.* For $x, y, z \in R$, we have $x \cdot (y + z) = x \cdot y + x \cdot z$ and $(x + y) \cdot z = x \cdot z + y \cdot z$.

[a] Again this has an implicit *closure* axiom

**Remark** (Notation)**.** For $x \in R$, we write $-x$ for its inverse under addition, and we abbreviate $x + (-y) = x - y$.

We note that in the definition we require $(R, +)$ to be an abelian group, but the fact that it is abelian is deducible from the other axioms. This means that if we talk about a ring being commutative, we must be referring to the multiplication.

**Definition 6.2** (Commutative Ring)

A ring $R$ is **commutative** if $x \cdot y = y \cdot x$ for all $x, y \in R$.

In this course, we will *only consider* commutative rings. So whenever we refer to an arbitrary ring, we will be referring to a commutative ring.[1]

We can define subrings in the natural way.

**Definition 6.3** (Subrings)

A subset $S \subset R$ is a **subring**, written $S \leq R$, if it is a ring under the same operations $+$ and $\cdot$, with the same identity elements 0 and 1.

**Example 6.4** (Examples of Rings and Subrings)

The following are examples of rings and subrings.

(i) We have the subrings $\mathbb{Z} \leq \mathbb{Q} \leq \mathbb{R} \leq \mathbb{C}$, under addition and multiplication as normal.

(ii) $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ is a ring, and $\mathbb{Z}[i] \leq \mathbb{C}$.

(iii) $\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\} \leq \mathbb{R}$.

(iv) $\mathbb{Z}\left[\frac{1}{p}\right] = \left\{\frac{m}{p^n} \mid m \in \mathbb{Z}, n \geq 0\right\} \leq \mathbb{Q}$.

(v) $\mathbb{Z}/n\mathbb{Z}$, the integers modulo $n$, is a ring of order $n$.

## §6.2 Constructing Rings

Similar to the direct product of groups, there is a natural way to combine rings.

**Definition 6.5** (Product Ring)

If $R$ and $S$ are rings, then their **product ring** $R \times S$ is a ring via $(r_1, s_1) + (r_2, s_2) = (r_1 + r_2, s_1 + s_2)$ and $(r_1, s_1) \cdot (r_2, s_2) = (r_1 \cdot r_2, s_1 \cdot s_2)$.

We have $0_{R \times S} = (0_R, 0_S)$ and $1_{R \times S} = (1_R, 1_S)$.

**Remark.** Note that $R \times \{0\}$ is *not* a subring of $R \times S$ – this doesn't have the same identity elements.

There are some other interesting ways to construct rings from other rings. First of all, we could consider functions from some set onto $R$.

[1] There are some notable non-commutative rings such as ones made up of matrices, but they will not be considered in this course.

**Definition 6.6**

If $R$ is a ring and $X$ is a set, then the set of all functions $X \to R$ is a ring under pointwise operations, $(f+g)(x) = f(x) + g(x)$, and $(f \cdot g)(x) = f(x) \cdot g(x)$.

Other interesting examples appear as subrings. For example, the continuous functions $\mathbb{R} \to \mathbb{R}$.

We can also construct the *ring of polynomials*

**Definition 6.7** (Ring of Polynomials)

Let $R$ be a ring and define $S$ to be the set of all sequences $(a_0, a_1, \dots)$ with $a_i \in R$ with $a_i = 0$ for all $i \geq N$ for some $N$. Then defining addition and multiplication as

$$\begin{aligned}(a_0, a_1, \dots) + (b_0, b_1, \dots) &= (a_0 + b_0, a_1 + b_1, \dots) \\ (a_0, a_1, \dots) \cdot (b_0, b_1, \dots) &= (c_0, c_1, \dots),\end{aligned}$$

where $c_n = \sum_{i=0}^n a_i b_{n-i}$ gives us $R[X]$, the **ring of polynomials in $X$ with coefficients in $R$**.

Note that if we define $X = (0, 1, 0, \dots)$, then $X^n = (\underbrace{0, 0, \dots, 0}_{n \text{ times}}, 1, 0, \dots)$, and

$$(a_0, a_1, \dots, a_n, 0, \dots) = a_n X^n + a_{n-1} X^{n-1} + \cdots + a_1 X + a_0.$$

This is why this is known as the polynomial ring.

You may wonder why we defined things in this way, and the answer is that we are not thinking about polynomials as functions, but as sequences of coefficients. To see why this matters, consider $R = \mathbb{Z}/p\mathbb{Z}$ with $p$ prime and $f(x) = X^p - X$. Then the function $x \mapsto f(x)$ is identically zero, but the polynomial $X^p - X$ is non-zero.

So we can restate our definition in a clearer and more general way.

**Definition 6.8** (Ring of Polynomials)

If $R$ is a ring then $R[X_1, \dots, X_n]$ is the *ring of polynomials* in $X_1, \dots, X_n$ with coefficients in $R$.

**Remark.** We can define this inductively as $R[X_1, \dots, X_n] = R[X_1, \dots, X_{n-1}][X_n]$ using our careful construction for $R[X]$ above.

Some extensions of this are the following rings.

**Definition 6.9** (Power Series and Laurent Polynomials)

The **power series ring** is the ring $R[[X]] = \{a_0 + a_1 X + a_2 X^2 + \cdots \mid a_i \in R\}$ and the **ring of Laurent polynomials** is $R[X, X^{-1}] = \{\sum_{i \in Z} a_i X^i \mid a_i \in R$, and only finitely many $a_i \neq 0\}$.

## §6.3 Units and Fields

We now introduce an important definition. Recall that not every element of a ring necessarily has an inverse under multiplication. When an element *does* have one, we call it a *unit*.

**Definition 6.10** (Unit)

An element $r \in R$ is a **unit** if it has an inverse under multiplication. That is, if there exists $s \in R$ such that $r \cdot s = 1$.

The units in a ring $R$ form a group $(R^\times, \cdot)$ under multiplication. For example, $\mathbb{Z}^\times = \{\pm 1\}$, and $\mathbb{Q}^\times = \mathbb{Q} \backslash \{0\}$.

When every non-zero element in a ring is a unit, we call that ring a *field*.

**Definition 6.11** (Field)

A **field** is a ring with $0 \neq 1$ such that every non-zero element is a unit.

Examples of fields are $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ and $\mathbb{Z}/p\mathbb{Z}$ where $p$ is prime.

**Remark.** Note that we require $0 \neq 1$ in the definition of a field. If we did not have this, then if $0 = 1$, $x = 1 \cdot x = 0 \cdot x = 0$, for all $x \in R$. So if a ring has $0 = 1$, then it is the trivial ring $R = \{0\}$. And we don't want this to be a field.

The notion of a ring allows us to do some interesting things.

**Lemma 6.12** (Polynomial Division Algorithm)

Let $f, g \in R[X]$. Suppose that the leading coefficient of $g$ is a unit. Then there exists $q, r \in R[X]$ such that $f(X) = q(X)g(X) + r(X)$, where $\deg(r) < \deg(g)$.

*Proof.* We induct on the degree of $f$. If the degree is $n$, then we write $f(X) = a_n X^n + \cdots + a_1 X + a_0$ with $a_n \neq 0$, and $g(X) = b_m X^m + \cdots + b_1 X + b_0$, where $b_m \in R^\times$.

If $n < m$, put $q = 0$ and $r = f$, and we are done. Otherwise, we put $f_1(X) = f(X) - a_n b_m^{-1} X^{n-m} g(X)$. Then the coefficient of $X^n$ is 0, and $\deg(f_1) < n$, and we apply the induction hypothesis to get $f_1(X) = q_1(X)g(X) + r(X)$, with $\deg(r) < \deg(g)$.

Then $f(X) = (q_1(X) + a_n b_m^{-1} X^{n-m})g(X) + r(X)$. □

**Remark.** If $R$ is a field, then we only need $g \neq 0$.

## §6.4 Homomorphisms and Ideals

As with groups, we are interested in the structure preserving maps between rings. Since a ring has two operations, we ask that both are preserved, and (importantly) that the multiplicative identity is sent to the multiplicative identity.

**Definition 6.13** (Ring Homomorphism)

Let $R$ and $S$ be rings. A function $\phi : R \to S$ is a **ring homomorphism** if

(i) $\phi(r_1 + r_2) = \phi(r_1) + \phi(r_2)$ for all $r_1, r_2 \in R$,

(ii) $\phi(r_1 \cdot r_2) = \phi(r_1) \cdot \phi(r_2)$ for all $r_1, r_2 \in R$,

(iii) $\phi(1_R) = 1_S$.

A bijective ring homomorphism is a **ring isomorphism**.

Note that $\phi(0_R) = 0_S$ comes for free, since $\phi$ is in particular a group homomorphism $(R,+) \to (S,+)$. The same reasoning gives us the following.

**Definition 6.14** (Kernel)

The **kernel** of a ring homomorphism $\phi : R \to S$ is $\ker(\phi) = \{r \in R \mid \phi(r) = 0\}$.

**Lemma 6.15**

A ring homomorphism $\phi$ is injective if and only if $\ker(\phi) = \{0\}$.

*Proof.* $\phi$ is, in particular, a homomorphism of additive groups, so this follows from the corresponding result for groups. □

Now here is where the theory of rings begins to diverge slightly from the theory of groups. In group theory, the subsets that could appear as kernels of homomorphisms were the normal subgroups. Let's think about what the kernel of a ring homomorphism looks like. It is certainly a subgroup under addition, but if $r \in R$ and $x \in \ker(\phi)$ then

$$\phi(rx) = \phi(r)\phi(x) = \phi(r) \cdot 0 = 0,$$

so the kernel 'absorbs' multiplication by arbitrary ring elements – something much stronger than being closed under multiplication. Subsets with this property are called *ideals*.

**Definition 6.16** (Ideal)

A subset $I \subseteq R$ is an **ideal**, written $I \trianglelefteq R$, if

(i) $I$ is a subgroup of $(R,+)$,

(ii) if $r \in R$ and $x \in I$, then $rx \in I$.

We say $I$ is **proper** if $I \neq R$.

By the discussion above, if $\phi : R \to S$ is a ring homomorphism then $\ker(\phi) \trianglelefteq R$.

**Remark.** If an ideal $I$ contains a unit $u$, then it contains $u^{-1}u = 1$, and then it contains $r \cdot 1 = r$ for every $r \in R$. So a proper ideal contains no units. In particular, note that a proper ideal is *not* a subring, since $1 \notin I$.

The most familiar example of an ideal comes from the integers.

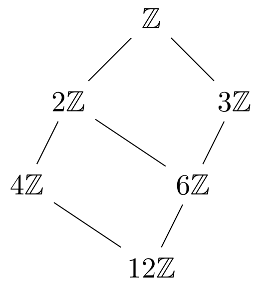
**Lemma 6.17** (Ideals of $\mathbb{Z}$)

The ideals of $\mathbb{Z}$ are exactly $n\mathbb{Z}$ for $n = 0, 1, 2, \ldots$.

*Proof.* Each $n\mathbb{Z}$ is easily checked to be an ideal. Conversely, let $I \trianglelefteq \mathbb{Z}$ be a non-zero ideal, and let $n$ be its smallest positive element. Then $n\mathbb{Z} \subseteq I$. If $m \in I$, then writing $m = qn + r$ with $0 \leq r < n$, we get $r = m - qn \in I$, so by minimality $r = 0$

and $m \in n\mathbb{Z}$. Thus $I = n\mathbb{Z}$. $\square$

Since $n\mathbb{Z} \subseteq m\mathbb{Z}$ if and only if $m \mid n$, the ideals of $\mathbb{Z}$ ordered by inclusion form precisely the divisibility lattice, drawn below for the ideals containing $12\mathbb{Z}$.




This example generalises: in any ring we can consider the set of 'multiples' of a fixed collection of elements.

**Definition 6.18** (Generated Ideals)
For $a \in R$, the **principal ideal** generated by $a$ is $(a) = \{ra \mid r \in R\}$. More generally, for $a_1, \dots, a_n \in R$ we write

$$(a_1, \dots, a_n) = \{r_1 a_1 + \cdots + r_n a_n \mid r_i \in R\},$$

the **ideal generated by** $a_1, \dots, a_n$.

For example, every ideal of $\mathbb{Z}$ is principal by the lemma above. Notice also that $(a) = R$ if and only if $a$ is a unit – a small observation that we will use constantly.

## §6.5 Quotient Rings

Just as we formed quotient groups from normal subgroups, we can form quotient rings from ideals. The additive structure is exactly the quotient group $(R, +)/(I, +)$, and the ideal property is precisely what makes multiplication of cosets well defined.

**Theorem 6.19** (Quotient Rings)
Let $I \trianglelefteq R$. Then the set $R/I$ of cosets of $I$ in $(R, +)$ is a ring, the **quotient ring**, with operations

$$(r_1 + I) + (r_2 + I) = (r_1 + r_2) + I,$$
$$(r_1 + I) \cdot (r_2 + I) = (r_1 r_2) + I,$$

and identities $0_{R/I} = 0 + I$ and $1_{R/I} = 1 + I$. Moreover the **quotient map** $\pi : R \to R/I$ with $r \mapsto r + I$ is a surjective ring homomorphism with kernel $I$.

*Proof.* From the theory of groups, addition is well defined and gives an abelian group. For multiplication, suppose $r_1 + I = r_1' + I$ and $r_2 + I = r_2' + I$, so $r_1' = r_1 + a_1$

and $r_2' = r_2 + a_2$ with $a_1, a_2 \in I$. Then

$$r_1' r_2' = r_1 r_2 + \underbrace{a_1 r_2 + r_1 a_2 + a_1 a_2}_{\in I},$$

so $r_1' r_2' + I = r_1 r_2 + I$, and multiplication of cosets is well defined. The ring axioms are all inherited directly from $R$, as is easily checked. The quotient map is then a surjective homomorphism by construction, and $\pi(r) = 0 + I \iff r \in I$. $\square$

So exactly as in the group case, ideals are precisely the kernels of ring homomorphisms. Let's look at some examples of quotient rings.

**Example 6.20** (Quotients of $\mathbb{Z}$)

The quotient of $\mathbb{Z}$ by the ideal $n\mathbb{Z}$ is the familiar ring $\mathbb{Z}/n\mathbb{Z}$ of integers modulo $n$.

**Example 6.21** ($\mathbb{C}[X]/(X)$)

Consider the ideal $(X) \trianglelefteq \mathbb{C}[X]$, the set of polynomials with zero constant term. Given $f(X) = a_n X^n + \cdots + a_1 X + a_0$, we have $f + (X) = a_0 + (X)$. So the map $\mathbb{C}[X]/(X) \to \mathbb{C}$ given by $f + (X) \mapsto f(0)$ is a well defined bijection, and it is easily checked to be a ring homomorphism. Thus $\mathbb{C}[X]/(X) \cong \mathbb{C}$.

**Example 6.22** ($\mathbb{R}[X]/(X^2+1)$)

Consider the ideal $(X^2+1) \trianglelefteq \mathbb{R}[X]$. Using the polynomial division algorithm, any $f \in \mathbb{R}[X]$ can be written $f(X) = q(X)(X^2+1) + r(X)$ with $\deg r < 2$, so every coset has a representative $a + bX$ with $a, b \in \mathbb{R}$, and it's straightforward to check this representative is unique. The bijection

$$\mathbb{R}[X]/(X^2+1) \longrightarrow \mathbb{C}, \quad a + bX + (X^2+1) \longmapsto a + bi$$

respects addition, and for multiplication note that $(a+bX)(c+dX) = ac + (ad + bc)X + bdX^2 \equiv (ac - bd) + (ad+bc)X \pmod{X^2+1}$, matching $(a+bi)(c+di) = (ac-bd) + (ad+bc)i$. So $\mathbb{R}[X]/(X^2+1) \cong \mathbb{C}$ – quotienting by $(X^2+1)$ amounts to 'adjoining a square root of $-1$'.

This last example illustrates an important idea – quotients of polynomial rings let us *build* rings in which some chosen polynomial has a root. We will return to this when we discuss fields and algebraic integers.

## §6.6 The Isomorphism Theorems for Rings

Each of the isomorphism theorems for groups has an analogue for rings, with normal subgroups replaced by ideals. The proofs consist of applying the group versions to the additive groups, and then checking everything respects multiplication.

**Theorem 6.23** (First Isomorphism Theorem)

Let $\phi : R \to S$ be a ring homomorphism. Then $\ker(\phi) \trianglelefteq R$, $\operatorname{img}(\phi) \leq S$ is a subring,

and
$$R/\ker(\phi) \cong \operatorname{img}(\phi).$$

*Proof.* We have already seen $\ker(\phi) \trianglelefteq R$. The image is an additive subgroup, it's closed under multiplication as $\phi(r_1)\phi(r_2) = \phi(r_1 r_2)$, and $1_S = \phi(1_R) \in \operatorname{img}(\phi)$, so $\operatorname{img}(\phi)$ is a subring of $S$.

Now let $K = \ker(\phi)$ and define $\Phi : R/K \to \operatorname{img}(\phi)$ by $r + K \mapsto \phi(r)$. By the first isomorphism theorem for groups, this is a well defined bijective homomorphism of additive groups. Also

$$\Phi((r_1+K)(r_2+K)) = \Phi(r_1r_2+K) = \phi(r_1r_2) = \phi(r_1)\phi(r_2) = \Phi(r_1+K)\Phi(r_2+K),$$

and $\Phi(1+K) = \phi(1_R) = 1_S$, so $\Phi$ is a ring isomorphism. $\square$

**Example 6.24**

Consider the 'evaluation at $i$' homomorphism $\phi : \mathbb{R}[X] \to \mathbb{C}$ given by $f \mapsto f(i)$. This is surjective, since $a + bX \mapsto a + bi$. Its kernel consists of real polynomials with $i$ as a root, and dividing any such polynomial by $X^2 + 1$ shows $\ker(\phi) = (X^2 + 1)$. So the first isomorphism theorem gives $\mathbb{R}[X]/(X^2 + 1) \cong \mathbb{C}$ immediately – much more efficiently than our direct verification before.

**Theorem 6.25** (Second Isomorphism Theorem)

Let $R \leq S$ be a subring and $J \trianglelefteq S$. Then $R \cap J \trianglelefteq R$, the set $R + J = \{r + a \mid r \in R, a \in J\}$ is a subring of $S$, and

$$R/(R \cap J) \cong (R + J)/J.$$

*Proof.* Let $\phi : R \to S/J$ be the composite of the inclusion $R \hookrightarrow S$ and the quotient map $S \to S/J$, so $\phi(r) = r + J$. This is a ring homomorphism, with

$$\ker(\phi) = \{r \in R \mid r + J = J\} = R \cap J \trianglelefteq R,$$

and $\operatorname{img}(\phi) = \{r+J \mid r \in R\} = (R+J)/J$, which is then a subring of $S/J$. The result follows by the first isomorphism theorem. (One can also check directly that $R + J$ is a subring of $S$ – it is an additive subgroup containing 1, and $(r_1 + a_1)(r_2 + a_2) = r_1r_2 + (r_1a_2 + r_2a_1 + a_1a_2) \in R + J$.) $\square$

For the third isomorphism theorem, we again start with a correspondence. If $I \trianglelefteq R$, then there is a bijection

$$\{\text{ideals of } R/I\} \longleftrightarrow \{\text{ideals of } R \text{ containing } I\},$$

given by $K \mapsto \{r \in R \mid r + I \in K\}$ and $J \mapsto J/I = \{r + I \mid r \in J\}$.

**Theorem 6.26** (Third Isomorphism Theorem)

Let $I \subseteq J$ be ideals of $R$. Then $J/I \trianglelefteq R/I$ and

$$(R/I) / (J/I) \cong R/J.$$

*Proof.* Consider $\phi : R/I \to R/J$ with $r+I \mapsto r+J$. This is well defined (as $I \subseteq J$), and is a surjective ring homomorphism with kernel $\{r + I \mid r \in J\} = J/I$. Apply the first isomorphism theorem. $\square$

Let's see a genuinely useful consequence of this machinery.

**Example 6.27** (The Characteristic of a Ring)

For any ring $R$, there is exactly one ring homomorphism $\iota : \mathbb{Z} \to R$. Indeed, we are forced to set $\iota(1) = 1_R$, and then $\iota(n) = 1_R + \cdots + 1_R$ ($n$ times) for $n \geq 0$, and $\iota(-n) = -\iota(n)$; one checks this is a ring homomorphism.

Now $\ker(\iota) \trianglelefteq \mathbb{Z}$, so $\ker(\iota) = n\mathbb{Z}$ for a unique $n \geq 0$, called the **characteristic** of $R$. By the first isomorphism theorem, every ring contains a copy of $\mathbb{Z}/n\mathbb{Z}$ (interpreting $\mathbb{Z}/0\mathbb{Z}$ as $\mathbb{Z}$).

For example, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ have characteristic 0, while $\mathbb{Z}/p\mathbb{Z}$ and $(\mathbb{Z}/p\mathbb{Z})[X]$ have characteristic $p$.

## §6.7 Integral Domains

Many familiar rings have the property that a product of non-zero elements is non-zero. This is such a useful property that rings with it get their own name.

**Definition 6.28** (Integral Domain)

An **integral domain** is a ring $R$ with $0 \neq 1$ such that for all $a, b \in R$,

$$ab = 0 \implies a = 0 \text{ or } b = 0.$$

A non-zero element $a \in R$ for which there is a non-zero $b$ with $ab = 0$ is called a **zero divisor** – so an integral domain is a non-trivial ring with no zero divisors.

**Example 6.29** (Examples of Integral Domains)

Every field is an integral domain: if $ab = 0$ and $a \neq 0$, then $b = a^{-1}(ab) = 0$. Also, any subring of an integral domain is an integral domain, so for example $\mathbb{Z} \leq \mathbb{Q}$ and $\mathbb{Z}[i] \leq \mathbb{C}$ are integral domains.

On the other hand, $\mathbb{Z} \times \mathbb{Z}$ is not an integral domain, since $(1, 0) \cdot (0, 1) = (0, 0)$, and $\mathbb{Z}/6\mathbb{Z}$ is not, since $2 \cdot 3 = 0$.

Working in an integral domain makes polynomials behave the way we would like them to.

**Lemma 6.30**

If $R$ is an integral domain, then $R[X]$ is an integral domain, and $\deg(fg) = \deg f +$

$\deg g$ for non-zero $f, g \in R[X]$.

*Proof.* Let $f, g \neq 0$ have leading coefficients $a_n \neq 0$ and $b_m \neq 0$. The coefficient of $X^{n+m}$ in $fg$ is $a_n b_m$, which is non-zero since $R$ is an integral domain. So $fg \neq 0$ and $\deg(fg) = n + m$. $\square$

**Lemma 6.31** (Roots of Polynomials)

Let $R$ be an integral domain and $f \in R[X]$ be non-zero. Then $f$ has at most $\deg(f)$ roots in $R$.

*Proof.* We induct on $\deg f$. If $\deg f = 0$ then $f$ is a non-zero constant, with no roots. Now suppose $a$ is a root of $f$. Dividing by the monic polynomial $X - a$, we can write $f(X) = q(X)(X - a) + r$ with $r$ constant, and evaluating at $a$ shows $r = 0$, so $f = q \cdot (X - a)$ with $\deg q = \deg f - 1$. Any other root $b \neq a$ of $f$ satisfies $q(b)(b - a) = 0$, so $q(b) = 0$ since $R$ is an integral domain. By induction $q$ has at most $\deg f - 1$ roots, so $f$ has at most $\deg f$ roots. $\square$

This innocent-looking fact has a rather striking consequence.

**Theorem 6.32**

Let $F$ be a field. Then any finite subgroup $G \leq (F^\times, \cdot)$ is cyclic.

*Proof.* $G$ is a finite abelian group. If $G$ is not cyclic, then by the classification of finite abelian groups (using the invariant factor form $G \cong C_{d_1} \times \cdots \times C_{d_t}$ with $d_1 \mid \cdots \mid d_t$ and $t \geq 2$), $G$ contains a subgroup isomorphic to $C_d \times C_d$ for some $d \geq 2$. But then the polynomial $X^d - 1 \in F[X]$ has at least $d^2 > d$ roots in $F$, contradicting the previous lemma. $\square$

**Corollary 6.33**

$(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic for any prime $p$.

We record one more useful fact about integral domains.

**Proposition 6.34** (Finite Integral Domains are Fields)

Any finite integral domain is a field.

*Proof.* Let $R$ be a finite integral domain and $0 \neq a \in R$. Consider the map $\phi : R \to R$ given by $x \mapsto ax$. If $\phi(x) = \phi(y)$ then $a(x - y) = 0$, so $x = y$, that is, $\phi$ is injective. Since $R$ is finite, $\phi$ is a bijection, so there is some $b \in R$ with $ab = 1$. Thus every non-zero element is a unit. $\square$

## §6.8 The Field of Fractions

The integers are not a field, but they sit naturally inside one – the rationals, whose elements are ratios of integers. This construction works in any integral domain.

**Theorem 6.35** (Field of Fractions)

Let $R$ be an integral domain. Then there is a field $F$, the **field of fractions** of $R$, which contains $R$ as a subring and in which every element can be written as $ab^{-1}$ with $a, b \in R$, $b \neq 0$.

*Proof.* We mimic the construction of $\mathbb{Q}$ from $\mathbb{Z}$. Consider the set $S = \{(a, b) \mid a, b \in R, b \neq 0\}$ of 'fractions', with the equivalence relation

$$(a, b) \sim (c, d) \iff ad = bc.$$

This is clearly reflexive and symmetric. For transitivity, if $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$, then $adf = bcf = bde$, so $d(af - be) = 0$, and since $d \neq 0$ and $R$ is an integral domain[a], $af = be$.

Write $\frac{a}{b}$ for the equivalence class of $(a, b)$, and let $F = S/\sim$ with operations

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \qquad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}.$$

Note $bd \neq 0$, and one checks these operations are well defined and make $F$ a ring, with $0_F = \frac{0}{1}$ and $1_F = \frac{1}{1}$. If $\frac{a}{b} \neq 0_F$ then $a \neq 0$, and $\frac{a}{b} \cdot \frac{b}{a} = 1_F$, so $F$ is a field. Finally, identifying $r \in R$ with $\frac{r}{1} \in F$ embeds $R$ as a subring of $F$, and $\frac{a}{b} = ab^{-1}$. $\square$

[a] This is where we need $R$ to be an integral domain – without it, transitivity genuinely fails.

**Example 6.36**

The field of fractions of $\mathbb{Z}$ is $\mathbb{Q}$, and the field of fractions of $\mathbb{C}[X]$ is the field $\mathbb{C}(X)$ of **rational functions** $\frac{f(X)}{g(X)}$ with $f, g \in \mathbb{C}[X]$ and $g \neq 0$.

## §6.9 Maximal and Prime Ideals

We now introduce two special classes of ideals, which correspond exactly to when the quotient ring is a field or an integral domain. First, a small but important observation.

**Lemma 6.37** (Fields Have No Interesting Ideals)

A non-trivial ring $R$ is a field if and only if its only ideals are $\{0\}$ and $R$.

*Proof.* Suppose $R$ is a field and $I \trianglelefteq R$ is non-zero. Then $I$ contains a non-zero element, which is a unit, so $I = R$.

Conversely, suppose the only ideals are $\{0\}$ and $R$, and let $0 \neq x \in R$. Then $(x) \neq \{0\}$, so $(x) = R$. In particular $1 \in (x)$, meaning $xy = 1$ for some $y$, so $x$ is a unit. $\square$

**Definition 6.38** (Maximal Ideal)

An ideal $I \trianglelefteq R$ is **maximal** if $I \neq R$ and the only ideals containing $I$ are $I$ and $R$.

**Proposition 6.39**

Let $I \trianglelefteq R$. Then $I$ is maximal if and only if $R/I$ is a field.

*Proof.* $R/I$ is a field if and only if its only ideals are $\{0\} = I/I$ and $R/I$. By the correspondence between ideals of $R/I$ and ideals of $R$ containing $I$, this happens if and only if the only ideals of $R$ containing $I$ are $I$ and $R$, that is, if and only if $I$ is maximal. $\square$

**Definition 6.40** (Prime Ideal)

An ideal $I \trianglelefteq R$ is **prime** if $I \neq R$ and whenever $ab \in I$, either $a \in I$ or $b \in I$.

**Example 6.41** (Prime Ideals of $\mathbb{Z}$)

The ideal $n\mathbb{Z} \trianglelefteq \mathbb{Z}$ is prime if and only if $n = 0$ or $n$ is a prime number. Indeed, for $n = p$ prime this is the classical fact that $p \mid ab$ implies $p \mid a$ or $p \mid b$, and $(0)$ is prime since $\mathbb{Z}$ is an integral domain. Conversely if $n = uv$ with $u, v > 1$ then $uv \in n\mathbb{Z}$ but $u, v \notin n\mathbb{Z}$.

**Proposition 6.42**

Let $I \trianglelefteq R$. Then $I$ is prime if and only if $R/I$ is an integral domain.

*Proof.* Unwinding definitions: $R/I$ is an integral domain if and only if $(a+I)(b+I) = 0 + I$ implies $a + I = 0 + I$ or $b + I = 0 + I$; that is, if and only if $ab \in I$ implies $a \in I$ or $b \in I$. (And $I \neq R$ corresponds to $0 \neq 1$ in $R/I$.) $\square$

**Corollary 6.43**

Every maximal ideal is prime.

*Proof.* If $I$ is maximal then $R/I$ is a field, hence an integral domain, hence $I$ is prime. $\square$

The converse fails: $(0) \trianglelefteq \mathbb{Z}$ is prime but not maximal. As an application of this circle of ideas, note that if $R$ is an integral domain of characteristic $n$, then $\mathbb{Z}/n\mathbb{Z}$ embeds in $R$, so $\mathbb{Z}/n\mathbb{Z}$ is an integral domain, and thus $n$ is $0$ or prime. So *the characteristic of an integral domain (in particular, of any field) is either* $0$ *or a prime number.*

# §7 Factorisation in Integral Domains

We now begin the study of factorisation, which will occupy us for some time. The story we are trying to generalise is the fundamental theorem of arithmetic: every integer factors into primes, essentially uniquely. It turns out that in a general integral domain this can fail in interesting ways, and understanding when it holds leads to a hierarchy of classes of rings.

*Throughout this section, $R$ is an integral domain.*

## §7.1 Prime and Irreducible Elements

We first need the right vocabulary for talking about divisibility.

**Definition 7.1** (Divisibility & Associates)

For $a, b \in R$, we say $a$ **divides** $b$, written $a \mid b$, if $b = ac$ for some $c \in R$. Equivalently, $(b) \subseteq (a)$.

Elements $a, b$ are **associates** if $a = bu$ for a unit $u$. Equivalently, $(a) = (b)$.

Associate elements are 'the same up to units' – for factorisation purposes they are interchangeable, just as $6 = 2 \cdot 3 = (-2)(-3)$ in $\mathbb{Z}$.

**Definition 7.2** (Irreducible & Prime Elements)

Let $r \in R$ be non-zero and not a unit. Then:

(i) $r$ is **irreducible** if $r = ab$ implies $a$ or $b$ is a unit;

(ii) $r$ is **prime** if $r \mid ab$ implies $r \mid a$ or $r \mid b$.

In $\mathbb{Z}$ these two notions coincide (both giving the usual prime numbers, up to sign), but in general they do not, and the distinction is the crux of this whole section. Note that both notions depend on the ambient ring: 2 is prime in $\mathbb{Z}$ but a unit in $\mathbb{Q}$, and $2X$ is irreducible in $\mathbb{Q}[X]$ but not in $\mathbb{Z}[X]$ (where 2 is not a unit).

Let us relate these notions to ideals.

**Lemma 7.3**

$(r) \trianglelefteq R$ is a prime ideal if and only if $r = 0$ or $r$ is prime.

*Proof.* Suppose $(r)$ is prime and $r \neq 0$. Since prime ideals are proper, $r$ is not a unit. If $r \mid ab$ then $ab \in (r)$, so $a \in (r)$ or $b \in (r)$, that is, $r \mid a$ or $r \mid b$. So $r$ is prime.

Conversely, $(0) = \{0\}$ is prime as $R$ is an integral domain. If $r$ is prime then $(r) \neq R$, and $ab \in (r)$ means $r \mid ab$, so $r \mid a$ or $r \mid b$, giving $a \in (r)$ or $b \in (r)$. □

**Lemma 7.4** (Primes are Irreducible)

If $r \in R$ is prime, then $r$ is irreducible.

*Proof.* Suppose $r = ab$. Then certainly $r \mid ab$, so without loss of generality $r \mid a$, say $a = rc$. Then $r = rcb$, so $r(1 - cb) = 0$, and since $r \neq 0$ and $R$ is an integral domain, $cb = 1$. So $b$ is a unit. □

The converse is *false* in general, and it's worth seeing this fail concretely.

**Example 7.5** (An Irreducible That Isn't Prime)
Consider the ring

$$R = \mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\} \leq \mathbb{C},$$

an integral domain (being a subring of $\mathbb{C}$). Define the **norm** $N : R \to \mathbb{Z}_{\geq 0}$ by $N(a + b\sqrt{-5}) = a^2 + 5b^2$, which is multiplicative: $N(zw) = N(z)N(w)$.

First, the units of $R$ are $\pm 1$: if $rs = 1$ then $N(r)N(s) = 1$, so $N(r) = 1$, forcing $r = \pm 1$.

Next, 2 is irreducible in $R$: if $2 = rs$ with neither a unit, then $N(r)N(s) = 4$ with $N(r), N(s) > 1$, so $N(r) = 2$ – but $a^2 + 5b^2 = 2$ has no integer solutions. Similarly 3, $1 + \sqrt{-5}$ and $1 - \sqrt{-5}$ are irreducible (there are no elements of norm 3 either).

Now observe

$$2 \cdot 3 = 6 = (1 + \sqrt{-5})(1 - \sqrt{-5}).$$

So $2 \mid (1 + \sqrt{-5})(1 - \sqrt{-5})$, but 2 divides neither factor (taking norms: $N(2) = 4 \nmid 6 = N(1 \pm \sqrt{-5})$). Hence 2 is irreducible but *not prime*.

Notice what we have really found: two genuinely different factorisations of 6 into irreducibles, not related by reordering and units. Unique factorisation fails in $\mathbb{Z}[\sqrt{-5}]$.

## §7.2 Principal Ideal Domains

**Definition 7.6** (Principal Ideal Domain)
An integral domain $R$ is a **principal ideal domain** (PID) if every ideal of $R$ is principal, that is, of the form $(a)$ for some $a \in R$.

We have already shown that $\mathbb{Z}$ is a PID. In a PID, the pathology of the previous example cannot occur.

**Proposition 7.7** (Irreducible $\iff$ Prime in a PID)
Let $R$ be a principal ideal domain. Then every irreducible element of $R$ is prime.

*Proof.* Let $r$ be irreducible and suppose $r \mid ab$ but $r \nmid a$; we must show $r \mid b$. Since $R$ is a PID, $(a, r) = (d)$ for some $d \in R$. Then $d \mid r$, say $r = cd$. As $r$ is irreducible, $c$ or $d$ is a unit.

If $c$ were a unit, then $(r) = (d) = (a, r) \ni a$, so $r \mid a$, a contradiction. So $d$ is a unit, and $(a, r) = R$. In particular $1 = sa + tr$ for some $s, t \in R$. Multiplying by $b$,

$$b = sab + trb,$$

and $r$ divides both terms on the right ($r \mid ab$ for the first). Hence $r \mid b$. □

**Lemma 7.8** (Irreducibles Generate Maximal Ideals in a PID)
Let $R$ be a PID and $r \in R$ be non-zero. Then $r$ is irreducible if and only if $(r)$ is maximal.

*Proof.* Suppose $r$ is irreducible, so $(r) \neq R$. If $(r) \subseteq J \trianglelefteq R$, then $J = (a)$ for some $a$ (as $R$ is a PID), and $r = ab$ for some $b$. Since $r$ is irreducible, either $a$ is a unit, giving $J = R$, or $b$ is a unit, making $a$ and $r$ associates, giving $J = (r)$. So $(r)$ is maximal.

Conversely if $(r)$ is maximal then $r$ is not a unit, and if $r = ab$ then $(r) \subseteq (a) \subseteq R$, so $(a) = (r)$ (making $b$ a unit) or $(a) = R$ (making $a$ a unit). So $r$ is irreducible. (This direction holds in any integral domain.) $\square$

Combining our results: in a PID, for non-zero $r$,

$$(r) \text{ maximal} \iff r \text{ irreducible} \iff r \text{ prime} \iff (r) \text{ prime}.$$

So in a PID, the non-zero prime ideals are exactly the maximal ideals.

## §7.3 Euclidean Domains

How do we actually *prove* that a ring is a PID? For $\mathbb{Z}$, the key was the division algorithm. Abstracting this gives the following definition.

**Definition 7.9** (Euclidean Domain)

An integral domain $R$ is a **Euclidean domain** (ED) if there is a function $\phi : R \setminus \{0\} \to \mathbb{Z}_{\geq 0}$, called a **Euclidean function**, such that

(i) if $a \mid b$ (with $a, b \neq 0$) then $\phi(a) \leq \phi(b)$,

(ii) if $a, b \in R$ with $b \neq 0$, then there are $q, r \in R$ with

$$a = bq + r, \quad \text{and either } r = 0 \text{ or } \phi(r) < \phi(b).$$

**Example 7.10**

$\mathbb{Z}$ is a Euclidean domain with $\phi(n) = |n|$, and for a field $F$, the polynomial ring $F[X]$ is a Euclidean domain with $\phi(f) = \deg f$ – property (ii) is exactly the polynomial division algorithm, which applies since every non-zero element of $F$ is a unit.

**Proposition 7.11** (ED implies PID)

Every Euclidean domain is a principal ideal domain.

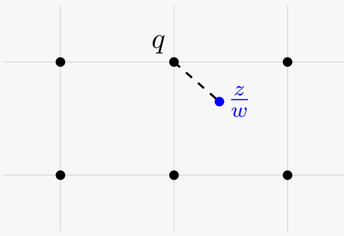
*Proof.* Let $R$ have Euclidean function $\phi$, and let $I \trianglelefteq R$ be a non-zero ideal. Choose $b \in I \setminus \{0\}$ with $\phi(b)$ minimal. Certainly $(b) \subseteq I$. Now take any $a \in I$, and write $a = bq + r$ with $r = 0$ or $\phi(r) < \phi(b)$. But $r = a - bq \in I$, so by minimality of $\phi(b)$ we must have $r = 0$. Hence $a \in (b)$, and $I = (b)$. $\square$

**Remark.** Only property (ii) was used here. The reason for including property (i) is that it lets us identify the units: they are exactly the non-zero elements with $\phi(u) = \phi(1)$, which is often handy in computations.

The most important new example is the ring of *Gaussian integers*.

**Lemma 7.12** ($\mathbb{Z}[i]$ is a Euclidean Domain)
The ring $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ is a Euclidean domain with Euclidean function $N(a + bi) = a^2 + b^2$.

*Proof.* The norm is multiplicative, $N(zw) = N(z)N(w)$, which gives property (i) at once. For property (ii), let $z, w \in \mathbb{Z}[i]$ with $w \neq 0$, and consider the complex number $\frac{z}{w} \in \mathbb{C}$. The elements of $\mathbb{Z}[i]$ form a unit square lattice in $\mathbb{C}$, so there is some $q \in \mathbb{Z}[i]$ with $\left|\frac{z}{w} - q\right| < 1$ (any point of $\mathbb{C}$ has distance at most $\frac{\sqrt{2}}{2} < 1$ from the nearest lattice point).




Now set $r = z - wq \in \mathbb{Z}[i]$. Then $z = wq + r$, and if $r \neq 0$,

$$N(r) = |z - wq|^2 = |w|^2 \left|\frac{z}{w} - q\right|^2 < |w|^2 = N(w),$$

as required. □

So $\mathbb{Z}[i]$ is a PID, and $F[X]$ is a PID for any field $F$. We will use both facts extensively. Here are two quick illustrations of the power of the $F[X]$ case.

**Example 7.13** (The Minimal Polynomial)
Let $A$ be an $n \times n$ matrix over a field $F$, and let

$$I = \{f \in F[X] \mid f(A) = 0\}.$$

Then $I$ is an ideal of $F[X]$ (differences and multiples of polynomials vanishing at $A$ vanish at $A$). Since $F[X]$ is a PID, $I = (f)$ for a single polynomial $f$, which we may scale to be monic – this is precisely the *minimal polynomial* of $A$, and we get for free that it divides every polynomial killing $A$.

**Example 7.14** (A Field of Order 8)
Let $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$ and $f(X) = X^3 + X + 1 \in \mathbb{F}_2[X]$. If $f$ factored non-trivially, it would have a linear factor and hence a root in $\mathbb{F}_2$, but $f(0) = f(1) = 1$. So $f$ is irreducible, and since $\mathbb{F}_2[X]$ is a PID, the ideal $(f)$ is maximal. Therefore

$$\mathbb{F}_2[X]/(X^3 + X + 1)$$

is a field. By the division algorithm each coset has a unique representative $aX^2 + bX + c$ with $a, b, c \in \mathbb{F}_2$, so this is a field with exactly 8 elements. This construction of finite fields is taken much further in Galois theory.

Not every PID-related fact extends to all polynomial rings, however.

**Example 7.15** ($\mathbb{Z}[X]$ is Not a PID)

Consider the ideal $I = (2, X) \trianglelefteq \mathbb{Z}[X]$, which consists of the polynomials with even constant term. Suppose $I = (f)$. Since $2 \in I$, we have $2 = fg$ for some $g$, so comparing degrees $f$ is a constant, and $f = \pm 1$ or $\pm 2$. If $f = \pm 1$ then $I = \mathbb{Z}[X]$, but $1 \notin I$. If $f = \pm 2$ then $X \notin (f)$, but $X \in I$. Either way we have a contradiction, so $I$ is not principal.

## §7.4 Unique Factorisation Domains

We can now formalise 'factorisation works like in $\mathbb{Z}$'.

**Definition 7.16** (Unique Factorisation Domain)

An integral domain $R$ is a **unique factorisation domain** (UFD) if

(i) every non-zero, non-unit element is a product of irreducibles, and

(ii) if $p_1 \cdots p_m = q_1 \cdots q_n$ with all $p_i, q_j$ irreducible, then $m = n$ and, after reordering, $p_i$ and $q_i$ are associates for each $i$.

The uniqueness half turns out to be equivalent to the prime/irreducible distinction collapsing.

**Proposition 7.17**

Let $R$ be an integral domain in which every non-zero, non-unit element is a product of irreducibles. Then $R$ is a UFD if and only if every irreducible element of $R$ is prime.

*Proof.* ($\Rightarrow$) Let $p$ be irreducible with $p \mid ab$, say $ab = pc$. Writing $a, b, c$ as products of irreducibles (or units), uniqueness of the factorisation of $ab$ forces $p$ to be an associate of one of the irreducible factors of $a$ or of $b$; hence $p \mid a$ or $p \mid b$.

($\Leftarrow$) Suppose every irreducible is prime, and $p_1 \cdots p_m = q_1 \cdots q_n$ with all factors irreducible, where without loss of generality $m \leq n$. Since $p_1$ is prime and divides $q_1 \cdots q_n$, it divides some $q_i$; reorder so $p_1 \mid q_1$. Then $q_1 = p_1 u$, and $u$ must be a unit as $q_1$ is irreducible and $p_1$ is not a unit. So $p_1, q_1$ are associates. Cancelling $p_1$ (valid in an integral domain) gives $p_2 \cdots p_m = u q_2 \cdots q_n$, and absorbing $u$ into $q_2$ we may induct. If $m < n$ we would eventually find that a product of irreducibles equals a unit, which is impossible, so $m = n$ and all pairs are associates. □

To show that PIDs are UFDs, the harder part is existence of factorisations, and for that we need a finiteness condition on chains of ideals – one which will reappear later when we study Noetherian rings.

**Definition 7.18** (Ascending Chain Condition)

A ring $R$ satisfies the **ascending chain condition** (ACC) if every increasing chain of ideals $I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$ is eventually constant, that is, there is some $N$ with

$I_n = I_{n+1}$ for all $n \geq N$. Rings satisfying the ACC are called **Noetherian**.

**Lemma 7.19** (PIDs Satisfy the ACC)
Every principal ideal domain satisfies the ascending chain condition.

*Proof.* Given a chain $I_1 \subseteq I_2 \subseteq \cdots$, the union $I = \bigcup_{i \geq 1} I_i$ is an ideal (a union of a chain of ideals is an ideal). Since $R$ is a PID, $I = (a)$ for some $a$, and $a \in I_N$ for some $N$. Then for all $n \geq N$,

$$(a) \subseteq I_N \subseteq I_n \subseteq I = (a),$$

so all inclusions are equalities. □

**Theorem 7.20** (PID implies UFD)
Every principal ideal domain is a unique factorisation domain.

*Proof.* We first show every non-zero, non-unit $x$ is a product of irreducibles. Suppose not. Then in particular $x$ is not itself irreducible, so $x = x_1 y_1$ with neither factor a unit, and at least one of them – say $x_1$ – is not a product of irreducibles. Note $(x) \subset (x_1)$, and this inclusion is strict since $y_1$ is not a unit. Repeating with $x_1$ in place of $x$, we build a strictly increasing chain

$$(x) \subset (x_1) \subset (x_2) \subset \cdots,$$

contradicting the ascending chain condition. So factorisations exist.

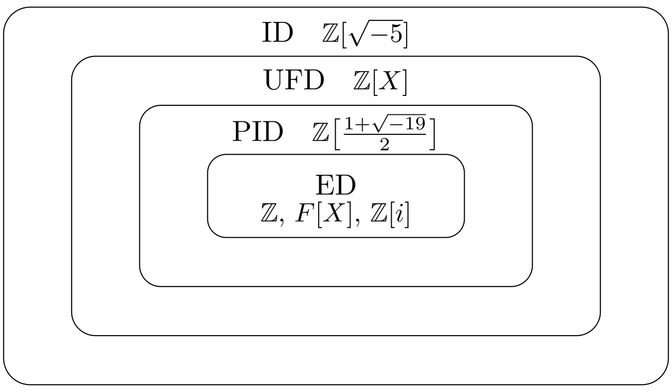
For uniqueness, by the previous proposition it suffices that every irreducible is prime – which we proved for PIDs earlier. □

## §7.5 The Hierarchy of Integral Domains

Assembling everything so far, we have a chain of implications

$$\text{ED} \implies \text{PID} \implies \text{UFD} \implies \text{ID},$$

and each implication is strict. The picture to have in mind is the following, with a witness sitting in each gap.



**Example 7.21** (Separating Examples)
Let's justify the diagram.

(i) $\mathbb{Z}$, $F[X]$ and $\mathbb{Z}[i]$ are Euclidean domains, as we have seen.

(ii) $\mathbb{Z}\left[\frac{1+\sqrt{-19}}{2}\right]$ is a PID but not a Euclidean domain. (Both facts require some work, and are beyond this course – this example is treated in Part II Number Fields.)

(iii) $\mathbb{Z}[X]$ is a UFD (as we will prove in the next section) but not a PID, since the ideal $(2, X)$ is not principal.

(iv) $\mathbb{Z}[\sqrt{-5}]$ is an integral domain but not a UFD, since 6 factors into irreducibles in two genuinely different ways.

And of course $\mathbb{Z}/4\mathbb{Z}$ fails even to be an integral domain, as $2 \cdot 2 = 0$.

## §7.6 Greatest Common Divisors

One pleasant consequence of unique factorisation is a good theory of greatest common divisors.

**Definition 7.22** (GCD and LCM)
Let $a_1, \ldots, a_n \in R$. An element $d \in R$ is a **greatest common divisor** of $a_1, \ldots, a_n$ if $d \mid a_i$ for all $i$, and any $d'$ with $d' \mid a_i$ for all $i$ satisfies $d' \mid d$.

Similarly, $m$ is a **least common multiple** if $a_i \mid m$ for all $i$, and every common multiple $m'$ satisfies $m \mid m'$.

Greatest common divisors need not exist in a general integral domain, but when they do they are unique up to associates – so we allow ourselves to speak of 'the' gcd.

**Proposition 7.23** (GCDs Exist in UFDs)
If $R$ is a UFD, then greatest common divisors and least common multiples always exist.

*Proof.* Write each $a_i = u_i \prod_j p_j^{n_{ij}}$, where the $p_j$ are pairwise non-associate irreducibles, $u_i$ are units, and $n_{ij} \geq 0$. We claim

$$d = \prod_j p_j^{m_j}, \quad \text{where } m_j = \min_i n_{ij},$$

is a greatest common divisor. Certainly $d$ divides each $a_i$. If $d' \mid a_i$ for all $i$, then writing $d'$ as a product of irreducibles, uniqueness of factorisation shows that (up to associates) $d' = w \prod_j p_j^{t_j}$ with $t_j \leq n_{ij}$ for every $i$, so $t_j \leq m_j$ and $d' \mid d$. The lcm is analogous, taking $\max_i n_{ij}$ instead. $\square$

# §8 Factorisation in Polynomial Rings

The goal of this section is the following theorem, together with some practical tools it generates for showing that specific polynomials are irreducible.

**Theorem 8.1** (Polynomial Rings over UFDs)
If $R$ is a unique factorisation domain, then $R[X]$ is a unique factorisation domain.

Note that this is exactly the sort of statement our hierarchy cannot give us directly: $\mathbb{Z}$ is a PID but $\mathbb{Z}[X]$ is not, so we cannot pass through principal ideals. Instead we will compare $R[X]$ with $F[X]$, where $F$ is the field of fractions of $R$. Since $F[X]$ is a Euclidean domain, it is a UFD, and the strategy is to transfer factorisations back and forth between $R[X]$ and $F[X]$.

*Throughout this section, $R$ is a UFD with field of fractions $F$, so $R[X] \leq F[X]$.*

## §8.1 Content and Gauss's Lemma

The bookkeeping device that makes the comparison work is the *content* of a polynomial.

**Definition 8.2** (Content & Primitive Polynomials)
The **content** of $f = a_0 + a_1X + \cdots + a_nX^n \in R[X]$ is

$$c(f) = \gcd(a_0, a_1, \ldots, a_n),$$

which is well defined up to associates (using that $R$ is a UFD). We say $f$ is **primitive** if $c(f)$ is a unit.

Any $f \in R[X]$ can be written $f = c(f)\, f_0$ with $f_0$ primitive – just pull out the gcd of the coefficients. The key technical fact is that primitivity is preserved by products.

**Lemma 8.3** (Products of Primitive Polynomials)
If $f, g \in R[X]$ are primitive, then so is $fg$. Moreover, for any $f, g \in R[X]$, the elements $c(fg)$ and $c(f)c(g)$ are associates.

*Proof.* Let $f = \sum_i a_iX^i$ and $g = \sum_i b_iX^i$ be primitive, and suppose $fg$ is not, so some irreducible (hence prime) $p \in R$ divides $c(fg)$. Since $f$ and $g$ are primitive, $p$ does not divide all the $a_i$, nor all the $b_i$. Let $k$ and $\ell$ be minimal with $p \nmid a_k$ and $p \nmid b_\ell$. The coefficient of $X^{k+\ell}$ in $fg$ is

$$\sum_{i+j=k+\ell} a_ib_j = \underbrace{(\cdots + a_{k-1}b_{\ell+1})}_{\text{divisible by } p} + a_kb_\ell + \underbrace{(a_{k+1}b_{\ell-1} + \cdots)}_{\text{divisible by } p},$$

where the flanking terms are divisible by $p$ by minimality of $k$ and $\ell$. Since $p$ divides the whole coefficient, $p \mid a_kb_\ell$, so $p \mid a_k$ or $p \mid b_\ell$ as $p$ is prime – a contradiction.

For the second part, write $f = c(f)f_0$ and $g = c(g)g_0$ with $f_0, g_0$ primitive. Then $fg = c(f)c(g)f_0g_0$ with $f_0g_0$ primitive, so taking contents gives $c(fg) = c(f)c(g)$ up to a unit. $\square$

**Corollary 8.4**

If $p \in R$ is prime in $R$, then $p$ is prime as an element of $R[X]$.

*Proof.* Note $R[X]^\times = R^\times$ (degrees add in an integral domain), so $p$ is a non-zero non-unit in $R[X]$. For $f \in R[X]$, we have $p \mid f$ in $R[X]$ if and only if $p \mid c(f)$ in $R$. So if $p \mid gh$, then $p \mid c(gh) = c(g)c(h)$ (up to units), hence $p \mid c(g)$ or $p \mid c(h)$, that is, $p \mid g$ or $p \mid h$ in $R[X]$. $\square$

The next lemma is the bridge between divisibility in $F[X]$ and in $R[X]$.

**Lemma 8.5** (Descending Divisibility)

Let $f, g \in R[X]$ with $g$ primitive. If $g \mid f$ in $F[X]$, then $g \mid f$ in $R[X]$.

*Proof.* Write $f = gh$ with $h \in F[X]$. Clearing denominators, there is a non-zero $a \in R$ with $ah \in R[X]$, and we can write $ah = c(ah)h_0$ with $h_0 \in R[X]$ primitive. Then

$$af = g \cdot ah = c(ah)\, gh_0,$$

and taking contents: $a \cdot c(f) = c(ah) \cdot c(gh_0) = c(ah)$ up to units, since $gh_0$ is a product of primitive polynomials, hence primitive. In particular $a \mid c(ah)$, so $h = \frac{c(ah)}{a} h_0 \in R[X]$, and $g \mid f$ in $R[X]$. $\square$

We can now prove the classical result relating irreducibility over $R$ and over the field of fractions.

**Lemma 8.6** (Gauss's Lemma)

Let $f \in R[X]$ be primitive. Then $f$ is irreducible in $R[X]$ if and only if $f$ is irreducible in $F[X]$.

*Proof.* ($\Rightarrow$) Since $f$ is irreducible and primitive, $\deg f > 0$, so $f$ is not a unit in $F[X]$. Suppose for contradiction that $f = gh$ with $g, h \in F[X]$ of positive degree. We claim we can arrange for $g$ to lie in $R[X]$ and be primitive. Indeed, choose $b \in R$ non-zero with $bg \in R[X]$, and write $bg = c(bg)g_0$ with $g_0$ primitive; then setting $\lambda = c(bg)b^{-1} \in F^\times$, we have $g = \lambda g_0$, and replacing $(g, h)$ with $(g_0, \lambda h)$ keeps $f = g_0(\lambda h)$. So assume $g \in R[X]$ is primitive. Then $g \mid f$ in $F[X]$, so by the previous lemma $g \mid f$ in $R[X]$, say $f = gh'$ with $h' \in R[X]$, where $\deg g, \deg h' > 0$. This contradicts irreducibility of $f$ in $R[X]$.

($\Leftarrow$) If $f = gh$ in $R[X]$ with $g, h$ non-units, then since $f$ is primitive neither factor can be constant, so both have positive degree and this also factors $f$ non-trivially in $F[X]$. $\square$

**Lemma 8.7**

Let $g \in R[X]$ be primitive. If $g$ is prime in $F[X]$, then $g$ is prime in $R[X]$.

*Proof.* $g$ is not a unit in $R[X]$ (it has positive degree, as primitive constants are units). If $g \mid f_1 f_2$ with $f_i \in R[X]$, then in $F[X]$ we get $g \mid f_1$ or $g \mid f_2$, and descending divisibility brings this back to $R[X]$. □

With these tools, the main theorem falls out.

*Proof that polynomial rings over UFDs are UFDs. Existence.* Let $f \in R[X]$ be a non-zero non-unit, and write $f = c(f)f_0$ with $f_0$ primitive. Since $R$ is a UFD, $c(f)$ factors into irreducibles of $R$, which remain irreducible in $R[X]$. For $f_0$: if it is not irreducible, then $f_0 = gh$ with $g, h$ non-units, and since $f_0$ is primitive both $g$ and $h$ have positive degree (and are primitive). By induction on degree, $f_0$ is a product of irreducibles.

*Uniqueness.* It suffices to show every irreducible $f \in R[X]$ is prime. Since $f$ is irreducible, it is either a constant or primitive. If $f$ is a constant, it is irreducible in $R$, hence prime in $R$ ($R$ being a UFD), hence prime in $R[X]$ by our corollary. If $f$ is primitive, then by Gauss's lemma $f$ is irreducible in $F[X]$; since $F[X]$ is a UFD, $f$ is prime in $F[X]$, and by the previous lemma $f$ is prime in $R[X]$. □

**Example 8.8**

$\mathbb{Z}[X]$ is a UFD, even though it is not a PID. Iterating the theorem, $R[X_1, \ldots, X_n] = R[X_1] \cdots [X_n]$ is a UFD whenever $R$ is – for instance $\mathbb{Z}[X_1, \ldots, X_n]$ and $F[X_1, \ldots, X_n]$ for a field $F$.

## §8.2 Eisenstein's Criterion

Gauss's lemma reduces irreducibility over $\mathbb{Q}$ to irreducibility over $\mathbb{Z}$, where we have more structure to exploit. The following criterion is the classic way to do so.

**Theorem 8.9** (Eisenstein's Criterion)

Let $R$ be a UFD and let $f(X) = a_n X^n + \cdots + a_1 X + a_0 \in R[X]$ be primitive with $\deg f = n \geq 1$. Suppose there is an irreducible $p \in R$ such that

(i) $p \nmid a_n$,

(ii) $p \mid a_i$ for all $i < n$,

(iii) $p^2 \nmid a_0$.

Then $f$ is irreducible in $R[X]$ (and hence in $F[X]$, by Gauss's lemma).

*Proof.* Suppose $f = gh$ with $g, h$ non-units. Since $f$ is primitive, $g$ and $h$ have positive degrees $k$ and $\ell$, with $k + \ell = n$. Write $g = \sum_{i=0}^{k} r_i X^i$ and $h = \sum_{i=0}^{\ell} s_i X^i$.

Since $p \nmid a_n = r_k s_\ell$, we have $p \nmid r_k$ and $p \nmid s_\ell$. Since $p \mid a_0 = r_0 s_0$ but $p^2 \nmid a_0$, exactly one of $r_0, s_0$ is divisible by $p$ – without loss of generality $p \mid r_0$ and $p \nmid s_0$.

Let $j \leq k$ be minimal with $p \nmid r_j$ (this exists as $p \nmid r_k$), so $j \geq 1$. Consider

$$a_j = r_0 s_j + r_1 s_{j-1} + \cdots + r_{j-1} s_1 + r_j s_0.$$

We have $j \le k < n$, so $p \mid a_j$, and $p$ divides each of the first $j$ terms by minimality of $j$. Hence $p \mid r_j s_0$, but $p \nmid r_j$ and $p \nmid s_0$ – a contradiction, as $p$ is prime ($R$ is a UFD). $\square$

**Example 8.10**

For a prime $p \in \mathbb{Z}$, the polynomial $X^n - p$ satisfies Eisenstein's criterion at $p$, so it is irreducible in $\mathbb{Z}[X]$, hence in $\mathbb{Q}[X]$. In particular $\sqrt[n]{p}$ is irrational for $n \ge 2$, and $\mathbb{Q}[X]/(X^n - p)$ is a field.

Sometimes the criterion doesn't apply directly, but does after a change of variable.

**Example 8.11** (Cyclotomic Polynomials of Prime Order)

Let $p$ be prime and consider

$$f(X) = X^{p-1} + X^{p-2} + \cdots + X + 1 = \frac{X^p - 1}{X - 1} \in \mathbb{Z}[X].$$

Eisenstein does not apply to $f$ directly (all the coefficients are 1). But substituting $X = Y + 1$,

$$f(Y+1) = \frac{(Y+1)^p - 1}{Y} = Y^{p-1} + \binom{p}{1} Y^{p-2} + \cdots + \binom{p}{p-2} Y + \binom{p}{p-1}.$$

Now $p \mid \binom{p}{i}$ for $1 \le i \le p-1$, and $\binom{p}{p-1} = p$ is not divisible by $p^2$, so Eisenstein applies at $p$ and $f(Y+1)$ is irreducible. Since $X \mapsto Y + 1$ is an isomorphism $\mathbb{Z}[X] \to \mathbb{Z}[Y]$, the polynomial $f(X)$ is irreducible in $\mathbb{Z}[X]$, and thus in $\mathbb{Q}[X]$.

**Example 8.12** (When Eisenstein Fails, Try Roots)

Consider $f(X) = X^3 + 2X + 5 \in \mathbb{Z}[X]$. No prime works for Eisenstein, but if $f$ factored in $\mathbb{Z}[X]$, then (being cubic) it would have a linear factor $X + a$ with $a \mid 5$, so an integer root among $\pm 1, \pm 5$. Checking, none of these are roots, so $f$ is irreducible in $\mathbb{Z}[X]$, and by Gauss's lemma also in $\mathbb{Q}[X]$. Hence $\mathbb{Q}[X]/(f)$ is a field.

# §9 The Gaussian Integers

The ring $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ deserves a section of its own – it is a Euclidean domain (with the norm $N(a+bi) = a^2 + b^2$ as Euclidean function), hence a PID and a UFD, and this structure turns out to encode genuinely arithmetic information about $\mathbb{Z}$: which integers are sums of two squares.

Since $N$ is multiplicative and $N(z) = 1 \iff z \in \{\pm 1, \pm i\}$, the units of $\mathbb{Z}[i]$ are exactly $\pm 1, \pm i$. Also note the handy fact that if $N(z)$ is prime in $\mathbb{Z}$, then $z$ is irreducible in $\mathbb{Z}[i]$, since any factorisation splits the norm.

**Example 9.1**

The prime 2 is *not* irreducible in $\mathbb{Z}[i]$: we have $2 = (1+i)(1-i)$, and neither factor is a unit ($N(1 \pm i) = 2$). Similarly $5 = (2+i)(2-i)$ is not irreducible. But 3 *is* irreducible: $N(3) = 9$, so a non-trivial factorisation $3 = zw$ would need $N(z) = 3$, and $a^2 + b^2 = 3$ has no integer solutions. The same argument shows 7 is irreducible.

There is a clear pattern here, and pinning it down is the main theorem of this section.

**Proposition 9.2**

Let $p \in \mathbb{Z}$ be a prime number. The following are equivalent:

(i) $p$ is not prime in $\mathbb{Z}[i]$;

(ii) $p = a^2 + b^2$ for some $a, b \in \mathbb{Z}$;

(iii) $p = 2$ or $p \equiv 1 \pmod 4$.

*Proof.* (i) $\Rightarrow$ (ii): Since $\mathbb{Z}[i]$ is a UFD, 'not prime' means 'not irreducible', so $p = zw$ with $z, w$ non-units. Then $p^2 = N(p) = N(z)N(w)$ with $N(z), N(w) > 1$, forcing $N(z) = p$. Writing $z = a + bi$ gives $p = a^2 + b^2$.

(ii) $\Rightarrow$ (iii): The squares modulo 4 are 0 and 1, so $a^2 + b^2 \not\equiv 3 \pmod 4$. Since $p$ is prime, either $p = 2$ or $p$ is odd with $p \equiv 1 \pmod 4$.

(iii) $\Rightarrow$ (i): We know $2 = (1+i)(1-i)$ is not prime, so suppose $p \equiv 1 \pmod 4$. The group $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p - 1$, and $4 \mid p - 1$, so there is an element $x$ of order 4. Then $x^2$ has order 2, and the only element of order 2 in the cyclic group is $-1$, so $x^2 \equiv -1 \pmod p$. Hence

$$p \mid x^2 + 1 = (x+i)(x-i) \quad \text{in } \mathbb{Z}[i].$$

But $p \nmid x \pm i$, since $\frac{x}{p} \pm \frac{1}{p} i \notin \mathbb{Z}[i]$. So $p$ is not prime in $\mathbb{Z}[i]$. $\square$

It is worth pausing here: the statement 'every prime $p \equiv 1 \pmod 4$ is a sum of two squares' is a theorem of Fermat, and makes no mention of complex numbers at all – yet the natural proof goes through the arithmetic of $\mathbb{Z}[i]$. Fermat's original approach was his method of *descent*: from any solution of $a^2 + b^2 \equiv 0 \pmod p$ he constructed a smaller multiple of $p$ expressible as a sum of two squares, descending until reaching $p$ itself. The Euclidean algorithm in $\mathbb{Z}[i]$ is doing exactly this descent for us, packaged as the statement that $\mathbb{Z}[i]$ is a PID.

We can now describe all the primes of $\mathbb{Z}[i]$.

**Theorem 9.3** (Primes in $\mathbb{Z}[i]$)

Up to associates, the primes of $\mathbb{Z}[i]$ are:

(i) elements $a + bi$ with $N(a + bi) = a^2 + b^2 = p$ a prime number with $p = 2$ or $p \equiv 1 \pmod 4$;

(ii) prime numbers $p \in \mathbb{Z}$ with $p \equiv 3 \pmod 4$.

*Proof.* Elements of type (i) are irreducible since their norm is prime, and elements of type (ii) are prime by the previous proposition. Conversely, let $z$ be prime in

$\mathbb{Z}[i]$, and note $\bar{z}$ is then also prime, with

$$N(z) = z\bar{z}$$

a factorisation into primes. Pick a prime number $p \in \mathbb{Z}$ dividing $N(z) \in \mathbb{Z}$.

If $p \equiv 3 \pmod 4$, then $p$ is prime in $\mathbb{Z}[i]$ and divides $z\bar{z}$, so $p \mid z$ or $p \mid \bar{z}$; in either case $z$ is an associate of $p$ (a prime dividing a prime), which is type (ii).

Otherwise $p = 2$ or $p \equiv 1 \pmod 4$, so $p = a^2 + b^2 = (a+bi)(a-bi)$ with both factors of norm $p$, hence prime. Then $(a+bi)(a-bi) \mid z\bar{z}$, and unique factorisation in $\mathbb{Z}[i]$ shows $z$ is an associate of $a+bi$ or $a-bi$, which is type (i). □

**Corollary 9.4** (Sums of Two Squares)

A positive integer $n$ is a sum of two squares if and only if every prime $p \equiv 3 \pmod 4$ appears in the factorisation of $n$ to an even power.

*Proof.* Note that $n$ is a sum of two squares if and only if $n = N(z)$ for some $z \in \mathbb{Z}[i]$. Factoring $z$ into primes of $\mathbb{Z}[i]$ and taking norms, $n$ is a product of norms of primes in $\mathbb{Z}[i]$, which by the classification are the primes $p \not\equiv 3 \pmod 4$ and the *squares* $p^2$ of primes $p \equiv 3 \pmod 4$. The stated condition describes exactly the integers expressible as products of such numbers. Conversely if the condition holds, writing each $p \not\equiv 3 \pmod 4$ as a norm and each $p^{2k}$ as $N(p^k)$, multiplicativity of $N$ expresses $n$ as a norm. □

**Example 9.5** (Writing 65 as a Sum of Two Squares)

We have $65 = 5 \cdot 13$, and both primes are 1 mod 4, so 65 should be a sum of two squares. Factoring in $\mathbb{Z}[i]$:

$$5 = (2+i)(2-i), \qquad 13 = (2+3i)(2-3i).$$

Grouping the four prime factors into conjugate pairs,

$$65 = N\big((2+i)(2+3i)\big) = N(1+8i) = 1^2 + 8^2,$$

but the alternative grouping gives

$$65 = N\big((2+i)(2-3i)\big) = N(7-4i) = 7^2 + 4^2.$$

So $65 = 1 + 64 = 49 + 16$ – and the two essentially different representations come precisely from the two ways of grouping the Gaussian primes.

## §10 Algebraic Integers

We now apply our factorisation theory to a class of complex numbers of central importance in number theory.

**Definition 10.1** (Algebraic Numbers and Algebraic Integers)

A number $\alpha \in \mathbb{C}$ is **algebraic** if it is a root of some non-zero polynomial in $\mathbb{Q}[X]$, and $\alpha$ is an **algebraic integer** if it is a root of some *monic* polynomial in $\mathbb{Z}[X]$.

For example, $\sqrt{2}$, $i$, and $\frac{-1+\sqrt{-3}}{2}$ (a root of $X^2+X+1$) are algebraic integers, while $\frac{1}{2}$ is algebraic but – as we shall see – not an algebraic integer.

For a subring $R \leq S$ and $\alpha \in S$, we write $R[\alpha]$ for the smallest subring of $S$ containing both $R$ and $\alpha$. Concretely, $R[\alpha] = \operatorname{img}(\phi)$, where $\phi : R[X] \to S$ is the **evaluation homomorphism** $g(X) \mapsto g(\alpha)$. This notation is consistent with our earlier usage of $\mathbb{Z}[i]$ and $\mathbb{Z}[\sqrt{-5}]$.

**Definition 10.2** (Minimal Polynomial)

Let $\alpha$ be an algebraic number, and consider the evaluation homomorphism $\phi : \mathbb{Q}[X] \to \mathbb{C}$, $g \mapsto g(\alpha)$. Since $\mathbb{Q}[X]$ is a PID, $\ker(\phi) = (f)$ for some $f$, which is non-zero as $\alpha$ is algebraic, and which we scale to be monic. This $f$ is the **minimal polynomial** of $\alpha$.

**Proposition 10.3**

The minimal polynomial $f$ of an algebraic number $\alpha$ is irreducible in $\mathbb{Q}[X]$, and $\mathbb{Q}[\alpha] \cong \mathbb{Q}[X]/(f)$ is a field.

*Proof.* By the first isomorphism theorem, $\mathbb{Q}[X]/(f) \cong \mathbb{Q}[\alpha] \leq \mathbb{C}$. A subring of a field is an integral domain, so $(f)$ is a prime ideal, and hence $f$ is prime, so irreducible. But then $(f)$ is a *maximal* ideal ($\mathbb{Q}[X]$ is a PID), so $\mathbb{Q}[X]/(f) \cong \mathbb{Q}[\alpha]$ is a field. □

The corresponding statement over $\mathbb{Z}$ requires more care, because $\mathbb{Z}[X]$ is not a PID – and this is exactly where Gauss's lemma earns its keep.

**Proposition 10.4**

Let $\alpha$ be an algebraic *integer* with minimal polynomial $f \in \mathbb{Q}[X]$. Then $f \in \mathbb{Z}[X]$, and $\ker(\theta) = (f)$, where $\theta : \mathbb{Z}[X] \to \mathbb{C}$ is the evaluation homomorphism at $\alpha$.

*Proof.* Choose $\lambda \in \mathbb{Q}^\times$ so that $\lambda f \in \mathbb{Z}[X]$ is primitive (clear denominators, then divide by the content). Then $\lambda f(\alpha) = 0$, so $\lambda f \in \ker(\theta)$.

We claim $\ker(\theta) = (\lambda f)$. Indeed, let $g \in \ker(\theta) \subseteq \mathbb{Z}[X]$. Then $g(\alpha) = 0$, so $f \mid g$ in $\mathbb{Q}[X]$, and hence $\lambda f \mid g$ in $\mathbb{Q}[X]$. Since $\lambda f$ is primitive, descending divisibility gives $\lambda f \mid g$ in $\mathbb{Z}[X]$, proving the claim.

Now, since $\alpha$ is an algebraic integer, there is a *monic* $g \in \mathbb{Z}[X]$ with $g(\alpha) = 0$, so $\lambda f \mid g$ in $\mathbb{Z}[X]$. Comparing leading coefficients, the leading coefficient of $\lambda f$ divides 1, so it is $\pm 1$; and as $f$ is monic, $\lambda = \pm 1$. Hence $f = \pm \lambda f \in \mathbb{Z}[X]$ and $\ker(\theta) = (f)$. □

Applying the first isomorphism theorem to $\theta$ gives presentations of our favourite rings:

$$\mathbb{Z}[X]/(X^2+1) \cong \mathbb{Z}[i], \qquad \mathbb{Z}[X]/(X^2-2) \cong \mathbb{Z}[\sqrt{2}], \qquad \mathbb{Z}[X]/(X^2+X+1) \cong \mathbb{Z}\left[\frac{-1+\sqrt{-3}}{2}\right],$$

and $\mathbb{Z}[X]/(X^n - p) \cong \mathbb{Z}[\sqrt[n]{p}]$ for a prime $p$.

**Corollary 10.5** (Rational Algebraic Integers)
If $\alpha \in \mathbb{Q}$ is an algebraic integer, then $\alpha \in \mathbb{Z}$.

*Proof.* If $\alpha \neq 0$ is rational, its minimal polynomial is $X - \alpha$, which by the previous proposition lies in $\mathbb{Z}[X]$. Hence $\alpha \in \mathbb{Z}$. (And $0 \in \mathbb{Z}$ trivially.) $\square$

This innocuous corollary is a genuinely useful tool – for instance, it is the cleanest way to see that $\frac{1}{2}$ is not an algebraic integer, and it underlies many irrationality proofs.

## §11 Noetherian Rings

We finish our study of rings by returning to the ascending chain condition, which we met while proving that PIDs are UFDs. It defines a class of rings large enough to contain almost everything we care about, but restrictive enough to do serious work.

**Definition 11.1** (Noetherian Ring)
A ring $R$ is **Noetherian** if every ascending chain of ideals $I_1 \subseteq I_2 \subseteq \cdots$ is eventually constant: there is $N$ such that $I_n = I_{n+1}$ for all $n \geq N$.

There is an equivalent characterisation which is often easier to verify and to use.

**Lemma 11.2** (Noetherian $\iff$ Finitely Generated Ideals)
A ring $R$ is Noetherian if and only if every ideal of $R$ is finitely generated.

*Proof.* ($\Leftarrow$) Let $I_1 \subseteq I_2 \subseteq \cdots$ be a chain and $I = \bigcup_i I_i$, an ideal. By assumption $I = (a_1, \ldots, a_k)$, and each $a_j$ lies in some $I_{n_j}$; taking $N = \max_j n_j$, all the generators lie in $I_N$, so for $n \geq N$,

$$I = (a_1, \ldots, a_k) \subseteq I_N \subseteq I_n \subseteq I,$$

and the chain is constant from $I_N$ onwards.

($\Rightarrow$) Suppose some ideal $J$ is not finitely generated. Pick $a_1 \in J$; then $(a_1) \neq J$, so pick $a_2 \in J \setminus (a_1)$, and inductively $a_{n+1} \in J \setminus (a_1, \ldots, a_n)$ – possible since $(a_1, \ldots, a_n)$ is finitely generated but $J$ is not. This produces a strictly increasing chain $(a_1) \subset (a_1, a_2) \subset \cdots$, contradicting the ACC. $\square$

In particular every PID is Noetherian – each ideal is generated by a single element. Fields are Noetherian for the same reason. The fundamental theorem about Noetherian rings is that the property passes to polynomial rings.

**Theorem 11.3** (Hilbert Basis Theorem)
If $R$ is Noetherian, then $R[X]$ is Noetherian.

*Proof.* Suppose for contradiction that $J \trianglelefteq R[X]$ is not finitely generated. Let $f_1 \in J$ have minimal degree; given $f_1, \dots, f_n$, let $f_{n+1} \in J \setminus (f_1, \dots, f_n)$ have minimal degree – such an element exists as $J$ is not finitely generated. Note the degrees $\deg f_1 \le \deg f_2 \le \cdots$ are non-decreasing by construction.

Let $a_i \in R$ be the leading coefficient of $f_i$. In $R$ we have the chain of ideals

$$(a_1) \subseteq (a_1, a_2) \subseteq (a_1, a_2, a_3) \subseteq \cdots,$$

which since $R$ is Noetherian is eventually constant – so there is $m$ with $a_{m+1} \in (a_1, \dots, a_m)$, say $a_{m+1} = \sum_{i=1}^m \lambda_i a_i$ with $\lambda_i \in R$. Now consider

$$g(X) = \sum_{i=1}^m \lambda_i X^{\deg f_{m+1} - \deg f_i} f_i \in (f_1, \dots, f_m),$$

which is engineered to have the same degree and the same leading coefficient as $f_{m+1}$. Then $f_{m+1} - g \in J$ with $\deg(f_{m+1} - g) < \deg f_{m+1}$, so by minimality of $\deg f_{m+1}$ we must have $f_{m+1} - g \in (f_1, \dots, f_m)$. But then $f_{m+1} = (f_{m+1} - g) + g \in (f_1, \dots, f_m)$, contradicting our choice of $f_{m+1}$. □

**Corollary 11.4**

$\mathbb{Z}[X_1, \dots, X_n]$ and $F[X_1, \dots, X_n]$ (for a field $F$) are Noetherian.

**Lemma 11.5** (Quotients of Noetherian Rings)

If $R$ is Noetherian and $I \trianglelefteq R$, then $R/I$ is Noetherian.

*Proof.* A chain of ideals $J_1' \subseteq J_2' \subseteq \cdots$ in $R/I$ corresponds, via the ideal correspondence, to a chain $J_1 \subseteq J_2 \subseteq \cdots$ of ideals of $R$ containing $I$, with $J_i' = J_i/I$. The latter chain stabilises since $R$ is Noetherian, so the former does too. □

So, for example, *any finitely generated ring* – any quotient of some $\mathbb{Z}[X_1, \dots, X_n]$ – is Noetherian. Let's see why one might care.

**Example 11.6** (Algebraic Varieties)

Let $R = \mathbb{C}[X_1, \dots, X_n]$, and let $\mathcal{F} \subseteq R$ be any (possibly infinite) set of polynomials, with solution set

$$V = \{(a_1, \dots, a_n) \in \mathbb{C}^n \mid f(a_1, \dots, a_n) = 0 \text{ for all } f \in \mathcal{F}\},$$

an **algebraic variety**. Let $I \trianglelefteq R$ be the ideal generated by $\mathcal{F}$. Since $R$ is Noetherian, $I = (g_1, \dots, g_r)$ is finitely generated, and $V$ is exactly the common zero set of $g_1, \dots, g_r$. So *every* system of polynomial equations, however large, is equivalent to a finite system.

**Example 11.7** (Non-Noetherian Rings)

The ring $\mathbb{Z}[X_1, X_2, \dots]$ of polynomials in countably many variables is not Noetherian: the chain $(X_1) \subset (X_1, X_2) \subset (X_1, X_2, X_3) \subset \cdots$ never stabilises. Note that this ring is an integral domain, and in fact a UFD – so 'UFD' does not imply 'Noetherian'.

More surprisingly, a subring of a Noetherian ring need not be Noetherian. Take

$$R = \{f \in \mathbb{Q}[X] \mid f(0) \in \mathbb{Z}\} \leq \mathbb{Q}[X].$$

Then $(X) \subset \left(\frac{X}{2}\right) \subset \left(\frac{X}{4}\right) \subset \cdots$ is a strictly increasing chain in $R$ (each inclusion is strict because 2 is not a unit in $R$), even though $\mathbb{Q}[X]$ itself is a Euclidean domain.

# §12 Modules

We now come to the final act of the course. A *module* is what you get when you take the definition of a vector space and allow the scalars to come from a ring rather than a field. This modest-sounding change has dramatic consequences – most of the comfortable facts of linear algebra (every vector space has a basis, spanning sets contain bases, and so on) fail for general modules – but for modules over sufficiently nice rings we will prove a powerful structure theorem, and from it deduce, in one stroke, the classification of finite abelian groups *and* the theory of canonical forms of matrices (including Jordan normal form).

## §12.1 Definitions and Examples

**Definition 12.1** (Module)
Let $R$ be a ring. An **$R$-module** is a triple $(M, +, \cdot)$ consisting of a set $M$, an operation $+ : M \times M \to M$, and a scalar multiplication $\cdot : R \times M \to M$, such that

(i) $(M, +)$ is an abelian group with identity $0 = 0_M$,

(ii) $(r_1 + r_2) \cdot m = r_1 \cdot m + r_2 \cdot m$,

(iii) $r \cdot (m_1 + m_2) = r \cdot m_1 + r \cdot m_2$,

(iv) $r_1 \cdot (r_2 \cdot m) = (r_1 r_2) \cdot m$,

(v) $1_R \cdot m = m$,

for all $r, r_1, r_2 \in R$ and $m, m_1, m_2 \in M$.

These are exactly the vector space axioms, with the field replaced by a ring. The real content of the definition is in the examples.

**Example 12.2** (Examples of Modules)
The following are all modules.

(i) If $R = F$ is a field, an $F$-module is precisely a vector space over $F$.

(ii) A $\mathbb{Z}$-module is precisely an abelian group. Indeed, given an abelian group $A$, the only possible $\mathbb{Z}$-action is

$$n \cdot a = \begin{cases} a + \cdots + a & (n \text{ times, if } n > 0), \\ 0 & (n = 0), \\ -(a + \cdots + a) & (|n| \text{ times, if } n < 0), \end{cases}$$

and this does define a module structure. So the theory of $\mathbb{Z}$-modules *is* the theory of abelian groups.

(iii) Let $F$ be a field, $V$ a vector space over $F$, and $\alpha : V \to V$ a linear map. Then $V$ becomes an $F[X]$-module, denoted $V_\alpha$, via

$$f(X) \cdot v = f(\alpha)(v).$$

Note that the module structure remembers $\alpha$, not just $V$ – different choices of $\alpha$ give genuinely different $F[X]$-modules on the same underlying space. This example will drive everything we do with canonical forms.

(iv) Any ring $R$ is an $R$-module over itself, and more generally $R^n$ is an $R$-module with coordinatewise operations.

(v) An ideal $I \trianglelefteq R$ is an $R$-module, as is the quotient ring $R/I$, with $r \cdot (s + I) = rs + I$.

(vi) If $\phi : R \to S$ is a ring homomorphism, then any $S$-module $M$ becomes an $R$-module via $r \cdot m = \phi(r) \cdot m$. In particular, a module over a ring is a module over any subring.

**Definition 12.3** (Submodule)

Let $M$ be an $R$-module. A subset $N \subseteq M$ is an ***R*-submodule**, written $N \leq M$, if $N$ is a subgroup of $(M, +)$ and $rn \in N$ for all $r \in R$, $n \in N$.

**Example 12.4**

Viewing $R$ as an $R$-module over itself, the submodules of $R$ are exactly the ideals of $R$. If $R = F$ is a field, submodules are exactly vector subspaces.

It is worth noticing what has happened here: subrings and ideals, which looked like two different generalisations of 'subgroup', are unified by the module point of view – and it is the ideal, not the subring, that is the module-theoretic notion.

**Definition 12.5** (Quotient Module)

Let $N \leq M$ be $R$-modules. The **quotient module** $M/N$ is the quotient abelian group with scalar multiplication $r \cdot (m + N) = rm + N$.

This is well defined precisely because $N$ is closed under scalar multiplication – note that, unlike for groups and rings, *every* submodule can be quotiented by; there is no analogue of normality to worry about.

## §12.2 Homomorphisms and the Isomorphism Theorems

**Definition 12.6** (Module Homomorphism)
Let $M, N$ be $R$-modules. A function $f : M \to N$ is an **$R$-module homomorphism** if

$$f(m_1 + m_2) = f(m_1) + f(m_2), \quad \text{and} \quad f(r \cdot m) = r \cdot f(m),$$

for all $m, m_1, m_2 \in M$ and $r \in R$. A bijective module homomorphism is an **isomorphism**.

When $R$ is a field, module homomorphisms are exactly linear maps. As always, we have kernels and images, and a family of isomorphism theorems, whose proofs at this point are routine: apply the group-theoretic versions to $(M, +)$, and check that all the maps involved respect scalar multiplication.

**Theorem 12.7** (First Isomorphism Theorem)
Let $f : M \to N$ be an $R$-module homomorphism. Then

$$\ker(f) = \{m \in M \mid f(m) = 0\} \le M, \qquad \operatorname{img}(f) = \{f(m) \mid m \in M\} \le N,$$

and $M/\ker(f) \cong \operatorname{img}(f)$, via the map $m + \ker(f) \mapsto f(m)$.

$$\begin{tikzcd}
M \arrow[r, "f"] \arrow[d, "\pi"'] & \operatorname{img}(f) \\
M/\ker(f) \arrow[ur, dashed, "\cong"'] &
\end{tikzcd}$$

**Theorem 12.8** (Second Isomorphism Theorem)
Let $A, B \le M$ be $R$-submodules. Then $A + B = \{a + b \mid a \in A, b \in B\}$ and $A \cap B$ are submodules of $M$, and

$$A/(A \cap B) \cong (A + B)/B.$$

**Theorem 12.9** (Third Isomorphism Theorem)
Let $N \le L \le M$ be $R$-submodules. Then $L/N \le M/N$, and

$$(M/N)\,/\,(L/N) \cong M/L.$$

There is also the familiar correspondence: submodules of $M/N$ biject with submodules of $M$ containing $N$.

**Remark.** Since vector spaces are modules, these specialise to facts from linear algebra – for instance, applying the first isomorphism theorem to a linear map $f : V \to W$ between finite dimensional spaces and taking dimensions gives

$$\dim V - \dim \ker(f) = \dim \operatorname{img}(f),$$

the rank-nullity theorem.

## §12.3 Finitely Generated Modules

We need a notion of 'finite dimensional' for modules.

**Definition 12.10** (Generation)

For $m \in M$, write $Rm = \{rm \mid r \in R\} \leq M$, the submodule **generated by** $m$. An $R$-module $M$ is **finitely generated** if

$$M = Rm_1 + Rm_2 + \cdots + Rm_n$$

for some $m_1, \ldots, m_n \in M$, that is, every element of $M$ is an $R$-linear combination $r_1 m_1 + \cdots + r_n m_n$.

The following reformulation is simple but surprisingly useful – it will be the opening move in the proof of the structure theorem.

**Lemma 12.11**

An $R$-module $M$ is finitely generated if and only if there is a surjective $R$-module homomorphism $f : R^n \to M$ for some $n$.

*Proof.* If $M = Rm_1 + \cdots + Rm_n$, define $f : R^n \to M$ by $f(r_1, \ldots, r_n) = r_1 m_1 + \cdots + r_n m_n$. This is an $R$-module homomorphism, and it is surjective by assumption.

Conversely, given such an $f$, let $e_i \in R^n$ be the standard 'basis' element with 1 in position $i$, and put $m_i = f(e_i)$. Any $m \in M$ equals $f(r_1, \ldots, r_n) = r_1 m_1 + \cdots + r_n m_n$ for some $r_i$, so the $m_i$ generate $M$. □

**Corollary 12.12**

Any quotient of a finitely generated module is finitely generated.

*Proof.* If $f : R^n \to M$ is surjective and $N \leq M$, then composing with the quotient map gives a surjection $R^n \to M/N$. □

**Example 12.13** (Submodules Need Not Be Finitely Generated)

In contrast with vector spaces (where subspaces of finite dimensional spaces are finite dimensional), a submodule of a finitely generated module can fail to be finitely generated. Take a non-Noetherian ring $R$ with an ideal $I$ that is not finitely generated – say $R = \mathbb{Z}[X_1, X_2, \ldots]$ and $I = (X_1, X_2, \ldots)$. Then $R$ is generated as an $R$-module by the single element 1, while its submodule $I$ is not finitely generated.

If $R$ is Noetherian, this cannot happen: submodules of finitely generated $R$-modules are finitely generated.

## §12.4 Torsion and Direct Sums

**Definition 12.14** (Torsion)

Let $M$ be an $R$-module. An element $m \in M$ is a **torsion element** if $rm = 0$ for some $r \neq 0$ in $R$. We say $M$ is a **torsion module** if every element is torsion, and **torsion-free** if 0 is the only torsion element.

**Example 12.15**

For $\mathbb{Z}$-modules (abelian groups), the torsion elements are exactly the elements of finite order; so a finite abelian group is a torsion $\mathbb{Z}$-module, and $\mathbb{Z}^n$ is torsion-free. Any module over a field is torsion-free. The $\mathbb{Z}$-module $\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ is neither torsion nor torsion-free.

**Definition 12.16** (Direct Sum)

Let $M_1, \ldots, M_n$ be $R$-modules. The **direct sum** $M_1 \oplus \cdots \oplus M_n$ is the set $M_1 \times \cdots \times M_n$ with componentwise addition and scalar multiplication.

For example, $R^n = R \oplus \cdots \oplus R$. We record one convenient compatibility between direct sums and quotients.

**Lemma 12.17**

Let $M = M_1 \oplus \cdots \oplus M_n$ and $N_i \leq M_i$ for each $i$. Then $N = N_1 \oplus \cdots \oplus N_n \leq M$ and

$$\frac{M_1 \oplus \cdots \oplus M_n}{N_1 \oplus \cdots \oplus N_n} \cong \frac{M_1}{N_1} \oplus \cdots \oplus \frac{M_n}{N_n}.$$

*Proof.* Apply the first isomorphism theorem to the surjective homomorphism $M \to \bigoplus_i M_i/N_i$ given by $(m_1, \ldots, m_n) \mapsto (m_1+N_1, \ldots, m_n+N_n)$, whose kernel is exactly $N$. □

## §12.5 Free Modules

We next isolate the modules that behave like vector spaces.

**Definition 12.18** (Independence)

Elements $m_1, \ldots, m_n \in M$ are **independent** if

$$r_1 m_1 + \cdots + r_n m_n = 0 \implies r_1 = \cdots = r_n = 0.$$

**Definition 12.19** (Freely Generating Sets & Free Modules)

A subset $S \subseteq M$ **generates $M$ freely** if $S$ generates $M$, and every function $\psi : S \to N$ into an $R$-module $N$ extends to an $R$-module homomorphism $\theta : M \to N$. (Such an extension is automatically unique, since $S$ generates $M$.) A module with a freely generating set is called **free**, and $S$ is a **free basis**.

**Proposition 12.20**

For a finite subset $S = \{m_1, \ldots, m_n\} \subseteq M$, the following are equivalent:

(i) $S$ generates $M$ freely;

(ii) $S$ generates $M$ and is independent;

(iii) every element of $M$ is *uniquely* expressible as $r_1 m_1 + \cdots + r_n m_n$;

(iv) the map $R^n \to M$, $(r_1, \ldots, r_n) \mapsto r_1 m_1 + \cdots + r_n m_n$, is an isomorphism.

*Proof.* We show (i) $\Rightarrow$ (ii); the equivalence of (ii), (iii) and (iv) is exactly as in linear algebra. Suppose $S$ generates $M$ freely but is not independent, say $\sum_i r_i m_i = 0$ with $r_j \neq 0$. Consider the function $\psi : S \to R$ with $\psi(m_j) = 1$ and $\psi(m_i) = 0$ for $i \neq j$, and extend it to a homomorphism $\theta : M \to R$. Then

$$0 = \theta(0) = \theta\left(\sum_i r_i m_i\right) = \sum_i r_i \theta(m_i) = r_j \neq 0,$$

a contradiction. For (iii) $\Rightarrow$ (i), define $\theta(r_1 m_1 + \cdots + r_n m_n) = \sum r_i \psi(m_i)$; uniqueness of the expression makes this well defined. ☐

So a finitely generated free module is just (isomorphic to) some $R^n$. But be warned – the intuition from vector spaces breaks down beyond this.

**Example 12.21** (Modules Are Not Vector Spaces)

The $\mathbb{Z}$-module $\mathbb{Z}/2\mathbb{Z}$ is finitely generated, but not free: every element $m$ satisfies $2m = 0$, so no non-empty subset is independent.

More subtly, $\{2, 3\}$ generates $\mathbb{Z}$ as a $\mathbb{Z}$-module (as $3 - 2 = 1$), but is not independent ($3 \cdot 2 - 2 \cdot 3 = 0$), and *no subset* of $\{2, 3\}$ is a free basis – $\{2\}$ and $\{3\}$ generate proper submodules. So a generating set need not contain a basis, in stark contrast with linear algebra.

One fact from linear algebra does survive, though it now requires genuine proof.

**Proposition 12.22** (Invariance of Rank)

Let $R$ be a non-trivial ring. If $R^n \cong R^m$ as $R$-modules, then $n = m$.

*Proof.* The idea is to reduce to the vector space case by quotienting. For an ideal $I \trianglelefteq R$ and an $R$-module $M$, define

$$IM = \{\textstyle\sum_i a_i m_i \mid a_i \in I,\ m_i \in M\} \leq M.$$

The quotient $M/IM$ is then naturally an $R/I$-module, via $(r + I)(m + IM) = rm + IM$ – this is well defined precisely because $IM$ absorbs multiplication by $I$.

Now let $I$ be a *maximal* ideal of $R$ (one exists, by a Zorn's lemma argument which we will not detail). Applying the above with $M = R^n$, note $IR^n = I^n$ (componentwise), so

$$(R/I)^n \cong R^n/IR^n \cong R^m/IR^m \cong (R/I)^m$$

as $R/I$-modules. But $R/I$ is a field, so these are vector spaces, and comparing dimensions gives $n = m$. ☐

## §12.6 Matrices over Euclidean Domains

We now begin working towards the main structure theorem. The engine of the proof is a normal form for *matrices* over a Euclidean domain, obtained by row and column operations – essentially Gaussian elimination, done without division.

*For the rest of this section, $R$ is a Euclidean domain with Euclidean function $\phi$, and we consider $m \times n$ matrices $A$ with entries in $R$.*

**Definition 12.23** (Elementary Operations)

The **elementary row operations** on a matrix over $R$ are:

(i) add $\lambda \in R$ times row $j$ to row $i$ (where $i \neq j$),

(ii) swap rows $i$ and $j$,

(iii) multiply row $i$ by a unit $u \in R^\times$.

Elementary column operations are defined analogously. Each is realised by multiplying on the left (respectively right) by an invertible matrix.

**Definition 12.24** (Equivalence)

Matrices $A$ and $B$ are **equivalent** if one can be obtained from the other by a sequence of elementary row and column operations; equivalently, $B = QAP$ for invertible $Q$ and $P$.

Notice what is *not* allowed: we cannot divide by non-units. The whole subtlety of the following algorithm is in working around this.

**Theorem 12.25** (Smith Normal Form)

Any $m \times n$ matrix $A$ over a Euclidean domain $R$ is equivalent to a diagonal matrix

$$\begin{pmatrix} d_1 & & & & \\ & \ddots & & & \\ & & d_t & & \\ & & & 0 & \\ & & & & \ddots \end{pmatrix},$$

with $d_1 \mid d_2 \mid \cdots \mid d_t$. The $d_i$ are called the **invariant factors** of $A$, and are unique up to associates.

*Proof.* If $A = 0$ we are done, so assume not, and by swapping rows and columns arrange $a_{11} \neq 0$. The plan is to repeatedly reduce $\phi(a_{11})$ until $a_{11}$ divides everything in sight; since $\phi$ takes values in $\mathbb{Z}_{\geq 0}$, this cannot go on forever.

*Step 1: clear the first row and column.* If $a_{11} \nmid a_{1j}$ for some $j$, write $a_{1j} = qa_{11} + r$ with $\phi(r) < \phi(a_{11})$ (and $r \neq 0$, else $a_{11} \mid a_{1j}$). Subtracting $q$ times column 1 from column $j$ and then swapping those columns puts $r$ in the top-left corner, strictly decreasing $\phi(a_{11})$. Do the same with rows if $a_{11} \nmid a_{i1}$. After finitely many iterations, $a_{11}$ divides every entry of the first row and column, and subtracting suitable multiples of the first row and column makes all those entries zero:

$$A \sim \begin{pmatrix} d & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & A' & \\ 0 & & & \end{pmatrix}, \quad d = a_{11}.$$

*Step 2: make $d$ divide everything.* If $d \nmid a_{ij}$ for some entry of $A'$, add row $i$ to row 1 – now the first row contains an entry not divisible by $d$, and we return to Step 1, which strictly decreases $\phi(a_{11})$ again. So after finitely many rounds, we reach the displayed form where moreover $d = d_1$ divides every entry of $A'$.

*Step 3: recurse.* Apply the algorithm to $A'$. The operations involved never touch the first row or column, and preserve the property that $d_1$ divides all entries (each entry of the new matrix is a combination of old ones). Inducting, $A$ is equivalent to $\operatorname{diag}(d_1, \ldots, d_t, 0, \ldots, 0)$ with $d_1 \mid d_2 \mid \cdots \mid d_t$.

Uniqueness of the $d_i$ is proven below, using the machinery of Fitting ideals. □

**Example 12.26**

Let us put the integer matrix $A = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}$ into Smith normal form. Working over $\mathbb{Z}$:

$$\begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} \xrightarrow{c_1 \mapsto c_1 + c_2} \begin{pmatrix} 1 & -1 \\ 3 & 2 \end{pmatrix} \xrightarrow{c_2 \mapsto c_2 + c_1} \begin{pmatrix} 1 & 0 \\ 3 & 5 \end{pmatrix} \xrightarrow{r_2 \mapsto r_2 - 3r_1} \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix},$$

which is in Smith normal form since $1 \mid 5$: the invariant factors are $1, 5$.

To prove uniqueness we introduce an invariant of the equivalence class of a matrix.

**Definition 12.27** (Minors and Fitting Ideals)

A $k \times k$ **minor** of $A$ is the determinant of a $k \times k$ submatrix of $A$ (obtained by deleting all but $k$ rows and $k$ columns). The **$k$-th Fitting ideal** $\operatorname{Fit}_k(A) \trianglelefteq R$ is the ideal generated by all $k \times k$ minors of $A$.

**Lemma 12.28**

Equivalent matrices have the same Fitting ideals.

*Proof.* By symmetry (transposing), it's enough to show row operations preserve $\operatorname{Fit}_k$. Swapping rows and scaling a row by a unit change each minor by at most a unit factor, so preserve the ideal they generate. Consider then the operation adding $\lambda$ times row $j$ to row $i$, giving matrix $A'$, and let $C'$ be a $k \times k$ submatrix of $A'$ with corresponding submatrix $C$ of $A$. If row $i$ is not involved, $C' = C$. If both rows $i$ and $j$ are involved, then $C'$ is obtained from $C$ by a row operation, so $\det C' = \det C$. If row $i$ is involved but row $j$ is not, expanding the determinant along the modified row gives

$$\det C' = \det C \pm \lambda \det D,$$

where $D$ is a $k \times k$ submatrix of $A$ (using row $j$ instead of row $i$, with rows reordered). In all cases $\det C' \in \mathrm{Fit}_k(A)$, so $\mathrm{Fit}_k(A') \subseteq \mathrm{Fit}_k(A)$, and since the operation is invertible, equality holds. □

*Proof of uniqueness.* Suppose the matrix $A$ is equivalent to $\mathrm{diag}(d_1, \ldots, d_t, 0, \ldots, 0)$ with $d_1 \mid \cdots \mid d_t$. A $k \times k$ minor of this diagonal matrix is non-zero only when it uses rows and columns with matching indices, in which case it is a product $d_{i_1} \cdots d_{i_k}$ with $i_1 < \cdots < i_k$. Each such product is divisible by $d_1 d_2 \cdots d_k$ (using the divisibility chain), and $d_1 \cdots d_k$ itself is a minor, so

$$\mathrm{Fit}_k(A) = (d_1 d_2 \cdots d_k).$$

Hence the products $d_1 \cdots d_k$ are determined by $A$ up to associates for each $k$, and dividing successive products recovers each $d_i$ up to associates. □

**Remark.** Fitting ideals also give a way to *compute* invariant factors without running the algorithm. In the example above, $\mathrm{Fit}_1(A) = (2, -1, 1, 2) = (1)$ gives $d_1 = \pm 1$, and $\mathrm{Fit}_2(A) = (\det A) = (5)$ gives $d_1 d_2 = \pm 5$, so $d_2 = \pm 5$.

## §12.7 The Structure Theorem

We now harvest the consequences of Smith normal form. The chain of reasoning runs as follows.

$$\begin{array}{c}
\boxed{M \text{ finitely generated, so } M \cong R^m/N} \\
\downarrow \\
\boxed{\text{generators of } N \text{ form the columns of a matrix } A} \\
\downarrow \\
\boxed{\text{put } A \text{ into Smith normal form } \mathrm{diag}(d_1, \ldots, d_t)} \\
\downarrow \\
\boxed{M \cong R/(d_1) \oplus \cdots \oplus R/(d_t) \oplus R^{m-t}}
\end{array}$$

We need two preliminary results to make this rigorous. The first says submodules of $R^m$ are manageable, and the second interprets row and column operations module-theoretically.

**Lemma 12.29** (Submodules of Free Modules)

Let $R$ be a Euclidean domain (or indeed any PID). Then any submodule $N \leq R^m$ is generated by at most $m$ elements.

*Proof.* We induct on $m$. Consider the ideal

$$I = \{r \in R \mid (r, r_2, \ldots, r_m) \in N \text{ for some } r_2, \ldots, r_m \in R\},$$

the set of possible first coordinates of elements of $N$ (an ideal, since $N$ is a submodule). As $R$ is a PID, $I = (a)$ for some $a$, and we may choose $n_1 = (a, a_2, \ldots, a_m) \in N$.

Now take any $x = (r_1, \ldots, r_m) \in N$. Then $r_1 = ra$ for some $r \in R$, and

$$x - rn_1 = (0, r_2 - ra_2, \ldots, r_m - ra_m) \in N \cap \left(\{0\} \times R^{m-1}\right) =: N'.$$

So $N = Rn_1 + N'$, where $N'$ is (identified with) a submodule of $R^{m-1}$, generated by at most $m-1$ elements $n_2, \ldots, n_m$ by induction. Then $n_1, n_2, \ldots, n_m$ generate $N$. □

**Theorem 12.30**

Let $R$ be a Euclidean domain and $N \leq R^m$ a submodule. Then there is a free basis $x_1, \ldots, x_m$ of $R^m$ such that $N$ is generated by $d_1x_1, d_2x_2, \ldots, d_tx_t$ for some $t \leq m$ and $d_1 \mid d_2 \mid \cdots \mid d_t$.

*Proof.* By the lemma, $N$ is generated by elements $y_1, \ldots, y_n \in R^m$ with $n \leq m$. Form the $m \times n$ matrix $A$ whose columns are the $y_i$, so that $N$ is generated by the columns of $A$.

Now consider what the elementary operations do to this data. A column operation replaces the generators $y_i$ by a different generating set for the *same* submodule $N$ (adding a multiple of one generator to another, swapping generators, or scaling by a unit). A row operation corresponds to changing the free basis of $R^m$: for instance, adding $\lambda$ times row $j$ to row $i$ amounts to expressing everything in the new basis $e_1, \ldots, e_j - \lambda e_i, \ldots, e_m$, which is again a free basis.

Putting $A$ into Smith normal form $\operatorname{diag}(d_1, \ldots, d_t, 0, \ldots)$ therefore produces a free basis $x_1, \ldots, x_m$ of $R^m$ (the accumulated row operations) together with generators $d_1x_1, \ldots, d_tx_t$ of $N$ (the non-zero columns). □

**Theorem 12.31** (Structure Theorem for Finitely Generated Modules over a Euclidean Domain)

Let $R$ be a Euclidean domain and $M$ a finitely generated $R$-module. Then

$$M \cong R/(d_1) \oplus R/(d_2) \oplus \cdots \oplus R/(d_t) \oplus R^k$$

for some $k \geq 0$ and non-zero $d_1 \mid d_2 \mid \cdots \mid d_t$. The $d_i$ are called the **invariant factors** of $M$, and (after discarding those $d_i$ which are units, for which $R/(d_i) = 0$) they are unique up to associates.

*Proof.* Since $M$ is finitely generated, there is a surjective homomorphism $f : R^m \to M$, and by the first isomorphism theorem $M \cong R^m/\ker(f)$. By the previous theorem, there is a free basis $x_1, \ldots, x_m$ of $R^m$ with $\ker(f)$ generated by $d_1x_1, \ldots, d_tx_t$, where $d_1 \mid \cdots \mid d_t$. Using this basis to identify $R^m$ with $R \oplus \cdots \oplus R$,

$$M \cong \frac{R \oplus \cdots \oplus R}{d_1R \oplus \cdots \oplus d_tR \oplus 0 \oplus \cdots \oplus 0} \cong R/(d_1) \oplus \cdots \oplus R/(d_t) \oplus R^{m-t},$$

using our lemma on quotients of direct sums. We omit the proof of uniqueness (which can be carried out by counting, for each prime $p \in R$ and each $j$, the 'sizes' of the modules $p^j M/p^{j+1}M$). □

**Remark.** The structure theorem holds more generally over any PID, with essentially the same statement. The proof runs along the same lines, but without a Euclidean function one needs a different induction to establish Smith normal form – one can measure the 'size' of a matrix entry by the number of irreducible factors, using unique factorisation.

Some immediate consequences are worth recording.

**Corollary 12.32**

Over a Euclidean domain, any finitely generated torsion-free module is free.

*Proof.* In the decomposition of the structure theorem, any factor $R/(d_i)$ with $d_i$ a non-zero non-unit contains torsion (the class of 1 is killed by $d_i$). Torsion-freeness forces all $d_i$ to be units, so $M \cong R^k$. □

**Example 12.33** (Computing with the Structure Theorem)

Let $G$ be the abelian group generated by elements $a$ and $b$ subject to the relations

$$2a + b = 0, \qquad -a + 2b = 0.$$

Formally, $G \cong \mathbb{Z}^2/N$, where $N \leq \mathbb{Z}^2$ is generated by $(2, 1)$ and $(-1, 2)$. These generators are the columns of

$$A = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix},$$

whose Smith normal form we computed: $\operatorname{diag}(1, 5)$. So in a suitable basis of $\mathbb{Z}^2$, $N$ is generated by $x_1$ and $5x_2$, and

$$G \cong \frac{\mathbb{Z} \oplus \mathbb{Z}}{\mathbb{Z} \oplus 5\mathbb{Z}} \cong \mathbb{Z}/5\mathbb{Z} \cong C_5.$$

## §12.8 Finitely Generated Abelian Groups

Since abelian groups are exactly $\mathbb{Z}$-modules and $\mathbb{Z}$ is a Euclidean domain, the structure theorem instantly classifies all finitely generated abelian groups.

**Theorem 12.34** (Classification of Finitely Generated Abelian Groups)

Let $G$ be a finitely generated abelian group. Then

$$G \cong C_{d_1} \times C_{d_2} \times \cdots \times C_{d_t} \times \mathbb{Z}^r$$

for some $r \geq 0$ and $d_1 \mid d_2 \mid \cdots \mid d_t$.

*Proof.* This is precisely the structure theorem for $R = \mathbb{Z}$, translated from module notation $(\mathbb{Z}/d\mathbb{Z}, \oplus)$ into group notation $(C_d, \times)$. $\square$

In particular, taking $G$ finite (so $r = 0$), this proves the fundamental theorem of finite abelian groups in its invariant factor form – the statement we promised back when we studied finite abelian groups. The prime power decomposition also generalises. Recall the key lemma was that $C_{mn} \cong C_m \times C_n$ for coprime $m, n$; here is the module version.

**Lemma 12.35** (Chinese Remainder Theorem)
Let $R$ be a PID and $a, b \in R$ with $\gcd(a, b)$ a unit. Then, as $R$-modules,

$$R/(ab) \cong R/(a) \oplus R/(b).$$

*Proof.* Since $R$ is a PID, $(a, b) = (d)$ where $d$ is a gcd of $a$ and $b$, so $(a, b) = R$ and we can write $1 = ra + sb$ for some $r, s \in R$.

Define $\psi : R \to R/(a) \oplus R/(b)$ by $\psi(x) = (x + (a), x + (b))$, an $R$-module homomorphism. Note

$$\psi(sb) = (sb + (a), sb + (b)) = (1 - ra + (a), 0 + (b)) = (1 + (a), 0 + (b)),$$

and similarly $\psi(ra) = (0 + (a), 1 + (b))$. Hence $\psi(x \cdot sb + y \cdot ra) = (x + (a), y + (b))$, so $\psi$ is surjective.

For the kernel: clearly $(ab) \subseteq \ker(\psi)$. Conversely if $x \in \ker(\psi)$ then $a \mid x$ and $b \mid x$, and writing $x = x(ra + sb) = r(ax) + s(bx)$, both terms are divisible by $ab$ (as $b \mid x$ in the first, $a \mid x$ in the second), so $x \in (ab)$. The first isomorphism theorem finishes the proof. $\square$

**Theorem 12.36** (Primary Decomposition)
Let $R$ be a Euclidean domain and $M$ a finitely generated $R$-module. Then

$$M \cong R/(p_1^{n_1}) \oplus \cdots \oplus R/(p_k^{n_k}) \oplus R^r$$

where the $p_i$ are primes of $R$ (not necessarily distinct) and $r \geq 0$.

*Proof.* By the structure theorem, it suffices to decompose each factor $R/(d)$. As $R$ is a UFD, write $d = up_1^{\alpha_1} \cdots p_s^{\alpha_s}$ with $u$ a unit and the $p_i$ pairwise non-associate primes. The prime powers $p_i^{\alpha_i}$ are pairwise coprime, so repeated application of the previous lemma gives

$$R/(d) \cong R/(p_1^{\alpha_1}) \oplus \cdots \oplus R/(p_s^{\alpha_s}).$$ $\square$

For $R = \mathbb{Z}$ this recovers the prime power form of the classification of finite abelian groups: every finite abelian group is a direct product of cyclic groups of prime power order.

## §12.9 Rational Canonical Form

We now turn to the second great application of the structure theorem: canonical forms for matrices. The bridge is the $F[X]$-module $V_\alpha$ from the start of this section – a vector space $V$ over a field $F$ together with a linear map $\alpha : V \to V$, with $X$ acting as $\alpha$.

**Lemma 12.37**

If $V$ is finite dimensional, then $V_\alpha$ is a finitely generated $F[X]$-module.

*Proof.* A basis $v_1, \ldots, v_n$ of $V$ as an $F$-vector space certainly generates $V_\alpha$ over the larger ring $F[X] \geq F$. $\square$

Since $F[X]$ is a Euclidean domain, the structure theorem applies to $V_\alpha$. To interpret the outcome we must understand what the cyclic modules $F[X]/(f)$ look like as 'vector space plus endomorphism' data.

**Example 12.38** (Companion Matrices)

Suppose $V_\alpha \cong F[X]/(f)$ where $f(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_0$ is monic. Then $1, X, X^2, \ldots, X^{n-1}$ is an $F$-vector space basis of $F[X]/(f)$ (by the division algorithm). The action of $\alpha$ is multiplication by $X$, which sends $X^i \mapsto X^{i+1}$ for $i < n-1$, and

$$X^{n-1} \longmapsto X^n = -a_{n-1}X^{n-1} - \cdots - a_1 X - a_0.$$

So with respect to this basis, $\alpha$ has matrix

$$C(f) = \begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{pmatrix},$$

the **companion matrix** of $f$.

**Theorem 12.39** (Rational Canonical Form)

Let $V$ be a finite dimensional vector space over a field $F$, and $\alpha : V \to V$ a linear map. Then

$$V_\alpha \cong F[X]/(f_1) \oplus F[X]/(f_2) \oplus \cdots \oplus F[X]/(f_t)$$

for monic polynomials $f_1 \mid f_2 \mid \cdots \mid f_t$, and with respect to a suitable basis of $V$, the map $\alpha$ has block diagonal matrix

$$\begin{pmatrix} C(f_1) & & & \\ & C(f_2) & & \\ & & \ddots & \\ & & & C(f_t) \end{pmatrix}.$$

*Proof.* $V_\alpha$ is a finitely generated module over the Euclidean domain $F[X]$, so the structure theorem gives

$$V_\alpha \cong F[X]/(f_1) \oplus \cdots \oplus F[X]/(f_t) \oplus F[X]^k$$

with $f_1 \mid \cdots \mid f_t$. Since $F[X]$ is infinite dimensional over $F$ while $V$ is finite dimensional, $k = 0$. Multiplying each $f_i$ by a unit, we may take them monic (and discard unit $f_i$). Choosing the companion basis in each summand and concatenating gives the block matrix. □

**Remark.** In matrix language: every square matrix over *any* field is similar to a block diagonal matrix of companion matrices with $f_1 \mid \cdots \mid f_t$. Reading off the diagonal: the *minimal polynomial* of $\alpha$ is $f_t$ (it kills every summand, and nothing smaller kills the last), and the *characteristic polynomial* is $f_1 f_2 \cdots f_t$ (the characteristic polynomial of $C(f)$ is $f$). In particular the minimal polynomial divides the characteristic polynomial – we have proven the Cayley-Hamilton theorem along the way.

**Example 12.40** ($2 \times 2$ Matrices)

Let $\dim V = 2$. Then $\sum_i \deg f_i = 2$, so either $t = 1$ with $\deg f_1 = 2$, or $t = 2$ with both linear. In the second case $f_1 \mid f_2$ forces $f_1 = f_2 = X - \lambda$, so $V_\alpha \cong F[X]/(X - \lambda) \oplus F[X]/(X - \lambda)$ and $\alpha = \lambda \cdot \mathrm{id}$ is scalar.

So: a non-scalar $2 \times 2$ matrix satisfies $V_\alpha \cong F[X]/(f)$ with $f$ its characteristic polynomial, and hence *two non-scalar* $2 \times 2$ *matrices are similar if and only if they have the same characteristic polynomial* – both being similar to the same companion matrix.

**Definition 12.41** (Annihilator)

The **annihilator** of an $R$-module $M$ is the ideal

$$\mathrm{Ann}_R(M) = \{r \in R \mid rm = 0 \text{ for all } m \in M\} \trianglelefteq R.$$

**Example 12.42**

For an ideal $I \trianglelefteq R$ we have $\mathrm{Ann}_R(R/I) = I$. For a finite abelian group $A$ viewed as a $\mathbb{Z}$-module, $\mathrm{Ann}_\mathbb{Z}(A) = (e)$ where $e$ is the exponent of $A$ – this finally delivers the proof (promised earlier) that the exponent is realised: in the invariant factor decomposition $A \cong C_{d_1} \times \cdots \times C_{d_t}$, the exponent is $d_t$, the order of an actual element. And for $V_\alpha$ as above, $\mathrm{Ann}_{F[X]}(V_\alpha) = (f_t)$ is generated by the minimal polynomial of $\alpha$.

## §12.10 Jordan Normal Form

Finally we specialise to $F = \mathbb{C}$, where the primes of the polynomial ring are as simple as possible.

**Lemma 12.43**

The irreducibles (equivalently, primes) in $\mathbb{C}[X]$ are exactly the linear polynomials $X - \lambda$ for $\lambda \in \mathbb{C}$, up to associates.

*Proof.* Linear polynomials are always irreducible. Conversely, by the fundamental theorem of algebra any non-constant $f \in \mathbb{C}[X]$ has a root $\lambda \in \mathbb{C}$, and then $X - \lambda$ divides $f$ (divide and inspect the remainder), so any $f$ of degree $\geq 2$ is reducible. □

So over $\mathbb{C}$, the primary decomposition of $V_\alpha$ is a direct sum of modules of the form $\mathbb{C}[X]/((X-\lambda)^n)$. Let's see what such a summand looks like.

**Example 12.44** (A Single Jordan Block)
Suppose $V_\alpha \cong \mathbb{C}[X]/((X-\lambda)^n)$, and take the $\mathbb{C}$-vector space basis

$$1,\ (X-\lambda),\ (X-\lambda)^2,\ \dots,\ (X-\lambda)^{n-1}.$$

The map $\alpha - \lambda \cdot \mathrm{id}$ acts as multiplication by $X - \lambda$, sending each basis vector to the next and the last to $(X-\lambda)^n = 0$. So in this basis $\alpha - \lambda\,\mathrm{id}$ is the shift matrix, and $\alpha$ has matrix

$$J_n(\lambda) = \begin{pmatrix} \lambda & & & \\ 1 & \lambda & & \\ & \ddots & \ddots & \\ & & 1 & \lambda \end{pmatrix},$$

an $n \times n$ **Jordan block**.[a]

[a] Depending on the choice of basis ordering, the 1's appear below or above the diagonal – both conventions are in use.

**Theorem 12.45** (Jordan Normal Form)
Let $V$ be a finite dimensional vector space over $\mathbb{C}$ and $\alpha : V \to V$ a linear map. Then

$$V_\alpha \cong \mathbb{C}[X]/((X-\lambda_1)^{n_1}) \oplus \cdots \oplus \mathbb{C}[X]/((X-\lambda_t)^{n_t})$$

with $\lambda_i \in \mathbb{C}$ not necessarily distinct, and with respect to a suitable basis, $\alpha$ has block diagonal matrix

$$\begin{pmatrix} J_{n_1}(\lambda_1) & & \\ & \ddots & \\ & & J_{n_t}(\lambda_t) \end{pmatrix}.$$

The Jordan blocks are unique up to reordering.

*Proof.* $V_\alpha$ is a finitely generated module over the Euclidean domain $\mathbb{C}[X]$, so the primary decomposition theorem applies; the primes of $\mathbb{C}[X]$ are the $X - \lambda$, and no free summand $\mathbb{C}[X]$ can appear by finite dimensionality. Choosing the basis of the previous example in each summand gives the block matrix. We will not prove uniqueness, but it follows by computing the dimensions of the generalised eigenspaces $\ker((\alpha - \lambda\,\mathrm{id})^m)$ for each $\lambda$ and $m$, which are determined by (and determine) the block sizes. □

**Remark** (Reading Off Data From the JNF)**.** Every square complex matrix is similar to one in Jordan normal form, and the form contains all similarity-invariant information:

(i) the eigenvalues of $\alpha$ are the $\lambda_i$ appearing in blocks;

(ii) the characteristic polynomial is $\prod_\lambda (X-\lambda)^{a_\lambda}$, where $a_\lambda$ is the sum of the sizes of the $\lambda$-blocks;

(iii) the minimal polynomial is $\prod_\lambda (X-\lambda)^{c_\lambda}$, where $c_\lambda$ is the size of the *largest* $\lambda$-block;

(iv) the dimension of the $\lambda$-eigenspace is the *number* of $\lambda$-blocks.

In particular, $\alpha$ is diagonalisable if and only if all Jordan blocks have size 1, if and only if the minimal polynomial has no repeated roots.

This is a satisfying place to stop and look back. The single idea of doing linear algebra over a ring – keeping track of *which* scalars we are allowed to divide by – has unified the classification of finite abelian groups, the theory of canonical forms, and (through the rings $\mathbb{Z}[i]$ and $F[X]$) a good deal of number theory. Each of these threads is picked up again in Part II, in Number Fields, Galois Theory and Representation Theory.